

Local Gambling Preference and Mortgage Misrepresentation

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September 2025

Abstract

During the housing boom preceding the Great Recession, higher household leverage was sometimes achieved through mortgage misrepresentation. This paper explores a large sample of mortgages originated between 2005 and 2007, focusing on the second-lien misreporting. We find that second-lien misreporting is more prevalent in areas with stronger gambling preferences, controlling for other local demographic characteristics and detailed loan and borrower characteristics. We also find that the effect is stronger when borrowers face higher incentives to misreport. In contrast, employing the FICO 620 threshold, we find little evidence that lenders' practices explain the relationship. Finally, we show that the negative performance consequences of second-lien misreporting are amplified in high-gambling-preference areas. We interpret our findings as evidence that borrowers' gambling preferences contributed to excessive household leverage in the years leading up to the financial crisis.

Keywords: Excessive leverage, Local gambling preference, Second-lien misrepresentation

JEL Codes: G4, G5, R2, R3

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1 Introduction

Household financial decisions have significant effects on the aggregate market (Campbell, 2006). A prominent example is the rapid expansion of U.S. household leverage, which played a central role in the 2008 financial crisis. From the expansion of credit supply to overly optimistic expectations about house prices, researchers find that households across all income groups took on excessive leverage in the mortgage market (Mian and Sufi, 2009; Adelino et al., 2016)¹. While much of this literature debates the underlying drivers, distorted beliefs versus misaligned incentives, relatively little attention has been paid to heterogeneity across households in such an aggressive borrowing environment. This paper provides empirical insights into which borrowers were more likely to take on excessive leverage.

In the years preceding the crisis, a striking example of excessive leverage was the misrepresentation of second liens, in which the existence of a simultaneous second lien was misreported at the time of first-lien origination (Piskorski et al., 2015; Griffin and Maturana, 2016). A standard first-lien mortgage already places a household in a leveraged position, and the addition of a second lien further increases leverage by reducing homeowner equity. Concealing this second lien at origination represents an extreme form of leverage taking because it allowed borrowers to appear less indebted than they actually were and made it easier for them to qualify for first-lien mortgages. This practice enabled households that otherwise lacked the income or credit capacity to obtain such large loans to overextend well beyond sustainable borrowing limits. Therefore, we focus on the setting of second-lien misrepresentation in this study.

In the context of second-lien misrepresentation, we focus on a specific behavioral trait: gambling preference, where people prefer small chances of large gains with large chances of small losses (Markowitz, 1952). The choice of gambling preference is motivated by its documented influence on a wide range of financial decision settings (e.g., Kumar et al., 2011;

¹Mian and Sufi (2009) show that subprime borrowers gained excessive access to mortgage credit that became disconnected from income growth, while Adelino et al. (2016) argue that middle- and high-income borrowers also increased leverage substantially and contributed to the rise in delinquencies during the crisis.

Han and Kumar, 2013) and by the finding of Christensen et al. (2018), who show that firms headquartered in areas with stronger gambling attitudes are more likely to misreport financial statements. We posit that similar mechanisms apply at the household level. From household’s perspective, second-lien misrepresentation resembles a gamble. If the gamble succeeds, misrepresentation provides attractive upsides such as homeownership and better loan terms². If it fails, the downside is loan denial and potential legal consequences, which are relatively limited compared to the potential gains³.

We measure second-lien misrepresentation and gambling preference following previous literature. We construct the second-lien misrepresentation measure following Piskorski et al. (2015), which compares loan-level mortgage data with borrowers’ credit report information. We follow Kumar (2009) and Kumar et al. (2011) to measure local gambling preference, using the ratio of Catholic residents to Protestant residents (CPRATIO) at the county level⁴. We choose this measure for two reasons. First, since the potential distribution of losses and gains for individual loans is not observable, we cannot directly apply existing theoretical frameworks in the housing market. Second, geographic measures are appropriate as people in the same area share social norms, while different areas often exhibit distinct cultural attitudes toward gambling. We interpret CPRATIO strictly as a regional cultural indicator of gambling preference that reflects social norms around speculative risk-taking in financial decisions, without inferring individual morality, personality traits, or religious affiliation⁵.

With these measures in place, we first investigate whether gambling preference helps explain the occurrence of mortgage misrepresentation. Comparing areas with different levels of local gambling preference, we find that second-lien misrepresentation is more likely to

²Emotionally, homeownership is associated with the American Dream in the U.S. (Clinton, 1995; Bush, 2003). Financially, with appropriate mortgage terms, homeownership brings large financial returns and helps households accumulate wealth (Bostic and Lee, 2008; Goodman and Mayer, 2018).

³Denial is common but the cost of loan application and the cost of switching lenders is relatively low; legal penalties (Henning, 2009; Federal Housing Finance Agency, 2023) were usually limited pre-crisis when law enforcement focused on fraud for profit rather than fraud for housing (FBI, 2006, 2007).

⁴We also employ alternative measures that do not rely on religiously constructed information in robustness check to mitigate the concern on broader religious difference.

⁵We also do not interpret our results as implying that members of particular religious groups are more likely to break the law than others. See Appendix for additional tests.

occur in counties with higher levels of local gambling preference, indicating it is an important driver. The economic magnitude also show that CPRATIO is a leading variable among all county-level variables.

To interpret this result from the perspective of borrowers, we test whether the relationship between gambling preference and misrepresentation is stronger when borrowers have greater incentives to misrepresent. We conduct subsample analyses by splitting the sample according to characteristics that reflect different incentive structures. The effect is indeed stronger when the property is owner-occupied (relative to non-primary) or when the loan is for purchase (relative to refinance), consistent with the interpretation from borrower perspective.

We also test whether the effect is in line with lenders' practices. Using the Keys et al. (2010) setting, where ease of securitization for low-documentation loans with FICO scores equal to or greater than 620 provides a shock to lenders but not borrowers, we apply a regression discontinuity design (RDD) to calculate jumps in number of loans and default rates, and compare these across high and low gambling preference areas using difference-in-discontinuities (diff-in-disc). Across all areas, the jump ratio in loans with misreported simultaneous seconds exceeds that of loans without seconds but is smaller than that of loans with correctly reported seconds, indicating lenders facilitate loans with simultaneous seconds but not specifically misrepresentation. The jump in default rates is statistically significant for loans with correctly reported simultaneous seconds but insignificant for those with misreported seconds, indicating that the lax screening effect is not larger for misreported seconds. Comparing high and low gambling preference areas, jumps in both number of loans and default rates for misreported seconds are smaller in high gambling preference areas, implying lenders there do not play a larger role in facilitating misrepresentation. Taken together, the positive correlation we found is less likely caused by lenders' practices.

Finally, we examine the outcomes of gambling preference when it is linked to mortgage misrepresentation. Given the strong correlation between gambling preference and misrep-

resentation, and the established association between misrepresentation and poor loan performance (Piskorski et al., 2015; Griffin and Maturana, 2016), we test whether gambling preference worsens outcomes through this channel. We find that loans with second-lien misrepresentation perform worse on average, and the effect is more pronounced in high gambling preference areas, indicating that gambling-motivated mortgage misrepresentation leads to poorer loan performance. A back-of-the-envelope calculation suggests that roughly \$0.8 billion of mortgage balances were affected through this channel, underscoring the economic importance of gambling attitudes in the mortgage market.

We conduct a series of robustness checks to reinforce our findings. First, to address the concern that CPRATIO may capture broader religious traits rather than gambling preference, we use alternative measures based on observed gambling activity and obtain consistent results. Second, to assess consistency with the credit expansion and belief distortion views of excessive leverage, we conduct subsample analyses and find patterns that align with these literatures, which also provide further support for interpreting gambling preference as a borrower-side channel. Third, to further support the idea that gambling leads to worse outcomes, we show that when misrepresentation is associated with high gambling preference, loan performance is poorer across all income and belief-distortion groups. Fourth, to ensure our results are not driven by a particular definition of default, we employ alternative, more or less restrictive measures and obtain similar outcomes. Fifth, to check that findings are not model-specific, we re-estimate using probit models and again find consistent results. Sixth, to strengthen causal inference, we apply the causal forest approach of Wager and Athey (2018), which yields similar conclusions. Finally, to confirm the robustness of our diff-in-disc results, we conduct multiple specification tests and find that the patterns in high and low gambling preference areas persist.

Our paper contributes to several strands of literature. First, it emphasizes the importance of heterogeneity in borrower characteristics, especially behavioral traits, in understanding household decisions to take on excessive leverage in the years leading up to the

financial crisis. Prior research highlights how credit supply expansion and distorted house price beliefs contributed to households taking on excessive leverage (e.g., Mian and Sufo, 2009; Adelino et al., 2018; Justiniano et al., 2019). While these studies emphasize external drivers of aggregate borrowing behavior, my paper shifts the focus to internal heterogeneity across households and provides evidence that borrowers reacted differently depending on their characteristics.

Second, our paper complements the literature by discussing where in the credit supply chain (from borrowers to MBS underwriters) second-lien misrepresentations took place. By studying privately securitized loans, Griffin and Maturana (2016) point out that lenders were likely aware of the misrepresentation, attributing such misreporting more to lenders than to underwriters. Conversely, Piskorski et al. (2015) show that underwriters could easily uncover the misrepresentation and also play an important role in this issue. Using both portfolio loans and securitized loans, Yavas and Zhu (2024) support the argument that second-lien misrepresentation occurs in the early stages of intermediation by lenders, while underwriters have limited but significant effects on reducing the occurrence of misrepresentation through screening. Our paper examines this issue from the borrower’s perspective, showing that borrowers significantly contribute to second-lien misrepresentation.

Third, our paper contributes to the literature on mortgage fraud by proposing a preference-based explanation. A substantial body of literature documents the prevalence of mortgage fraud and its economic impact (Elul et al., 2010; Piskorski et al., 2015; Ambrose et al., 2016; Griffin and Maturana, 2016; Mian and Suf, 2017; Kruger and Maturana, 2021). While previous studies have considered associated loan characteristics and household features, the role of preference in the decision-making process remains unclear. From a cultural perspective, Conklin et al. (2022) provide empirical evidence that religiosity helps constrain fraudulent activity but do not distinguish between ethics and risk channels. Bypassing the issue of differentiating ethics and risk preference, we show that gambling preference⁶ influences mortgage

⁶See Kumar (2009), Kumar et al. (2011), Kumar et al. (2016) for how to differentiate gambling preference from ethics.

applications.

Fourth, our paper builds on the empirical literature linking gambling preferences with investment decisions. Prior research has examined the effect of gambling preference on the stock market (Barberis and Huang, 2008; Kumar, 2009; Kumar et al., 2011; Barberis et al., 2016, 2021), bond market (Benartzi and Thaler, 1995), and option market (Baele et al., 2019). Additionally, literature explores its impact on corporate policy decisions (Kumar et al., 2011; Chen et al., 2014) and fund strategies (Shu et al., 2012). To the best of our knowledge, we are the first to apply the concept of gambling preference to the mortgage market in the context of household investment decision-making. This is important because existing literature on behavioral biases in the housing market predominantly centers on pricing dynamics and speculative bubbles from the investment perspective (Lux, 1995; Genesove and Mayer, 2001; Farlow, 2004; Shiller, 2005; Farlow, 2013), whereas relatively little attention has been paid to behavioral influences on mortgage choices, which is the financing side of housing decisions.

The rest of the paper is organized as follows. Section 2 describes the data and variables. In Section 3, we examine the effect of gambling preference on second-lien misrepresentation. In Section 4, we assess whether the effect of gambling preference aligns with borrower behavior or lender practice. Section 5 investigates the outcomes of gambling preference-related misrepresentation. Robustness checks are conducted in Section 6, and conclusions are presented in Section 7.

2 Data and Summary Statistics

Our sample comprises three main groups of data: loan-related records, religiosity data, and demographic information, covering the period from 2005 to 2007. These data are sourced from various datasets. Loan-related records include loan-level mortgage data from BlackBox Logic (now part of Moody’s) and borrower-level credit report information from Equifax. Religiosity data, which provides county-level information on prevalent religious adherence,

is obtained from the American Religion Data Archive (ARDA). Demographic information at the county and zipcode levels, such as age and income, is sourced from the U.S. Census Bureau. Additionally, we collect house price data from the Federal Housing Finance Agency (FHFA) and unemployment data from U.S. Bureau of Labor Statistics.

2.1 Second-lien Misrepresentation Measure

Second-lien misrepresentation occurs when a first-lien loan backed by property is reported as having no associated higher lien but is actually financed with a simultaneously originated second mortgage identified by credit bureau data. This misrepresentation allows the borrower to take on additional debt, reducing their incentive to repay the loans and making the initial debt riskier. We identify second-lien misrepresentation by comparing loan-level mortgage data from BlackBox Logic with borrower-level credit report information from Equifax. Following the procedure in Piskorski et al. (2015), we first focus on first-lien loans with a merge confidence interval greater than or equal to 0.89. Second, we select loans with a new second-lien originating within one month before or after the first lien⁷. This filter, using credit information, retains first-lien loans that truly have a simultaneous second lien. Third, to identify misreported second-lien loans, we require the loan to have a non-missing reported cumulative loan-to-value (CLTV) ratio within 1% of its loan-to-value (LTV) ratio. The small difference between the reported CLTV and LTV indicates that the borrower reports no simultaneous second lien.

2.2 Gambling Preference and Religiosity

Following Kumar et al. (2011), we construct county-level gambling preference based on religious data, specifically the ratio of Catholic residents to Protestant residents (CPRATIO).

⁷Piskorski et al. (2015) uses a 45-day range, while Zhang et al. (2024) uses a one-month range. Since Equifax data identifies the close date of the second lien at the month level, we conservatively use a one-month range. A 45-day range can be achieved by assuming the close date is in the middle of the month. We also checked the 45-day range and found minor differences in the results.

Although direct measures of local gambling preference are unavailable⁸, we can infer the propensity by examining the proportion of different religious populations, which have distinct attitudes towards gambling according to their religious views. In the U.S., two widespread religions with differing views on gambling are Catholicism and Protestantism. While Protestant churches generally oppose gambling, Catholic churches maintain a tolerant attitude towards moderate levels of gambling. These differing views between the two religions are empirically supported to extend to financial markets (Kumar, 2009; Kumar et al., 2011; Han and Kumar, 2013; Chen et al., 2014). Therefore, regions with higher Catholic-Protestant ratios exhibit stronger gambling propensities.

To use CPRATIO for potential causal inference, we need to control for religiosity in the county. According to Kumar et al. (2011) and Chen et al. (2014), religiosity should be considered because risk aversion increases with religiosity, regardless of the type of religion (Hilary and Hui, 2009). Including the overall level of religiosity in the county as a control ensures that our local gambling preference proxy is independent of religion-induced risk aversion. Additionally, religiosity plays a significant role in deterring various types of mortgage misrepresentation (Conklin et al., 2022) from the perspective of social norms and ethical behavior. Therefore, we follow Kumar et al. (2011) to construct the religiosity measure.

We use the dataset "Longitudinal Religious Congregations and Membership File, 1980-2010 (County Level)" from ARDA to capture county-level geographical variation in religious composition. We sum the number of adherents in the religious traditional categories Evangelical Protestant, Mainline Protestant, and Black Protestant for the Protestant population and the number of adherents in the category Catholic for the Catholic population. We use the sum of adherents in all categories for the total religious adherents in the county. Since the data is available for each decade from 1980 to 2010, we follow previous studies to linearly interpolate the data to obtain values for missing years (Alesina and La Ferrara, 2000; Hilary

⁸Gambling industry employment and establishment counts can serve as alternative measures of local gambling preference but have several limitations. We therefore use them only as robustness checks. See Appendix A for further discussion.

and Hui, 2009; Kumar et al., 2011; Chen et al., 2014), and then calculate the CPRATIO (i.e., Catholic population to Protestant population) and REL (i.e., total religious adherents to total population) for each year.

2.3 Geographic Controls

In addition to religiosity, variations in religion-induced gambling preferences may also correlate with other geographic characteristics (Kumar et al., 2011; Chen et al., 2014; Conklin et al., 2022). To help establish causality, we control for the following factors (Kumar, 2009; Kumar et al., 2011; Chen et al., 2014). The U.S. Census Bureau provides a rich set of demographic data. We employ county-level information in 2000 on education, marriage, living area, population, age, male-female ratio, and minority proportion. Since income plays an important role not only in potential correlation with religious distribution but also in household mortgage decisions, we use an even smaller cluster level, the zip code level, for control variables. We utilize county-level monthly data of unemployment rate from U.S. Bureau of Labor Statistics. Additionally, we collect annual house price indices (HPI) at the zip code level to calculate house price appreciation, which is also a crucial determinant in household mortgage decisions.

2.4 Mortgage Microdata

In addition to the second-lien misrepresentation measure, we also obtain other loan-level mortgage data from BlackBox Logic. The database includes a comprehensive set of loan characteristics at origination, such as the loan interest rate, borrower’s FICO credit score, initial loan balance, loan-to-value ratio, amortization type (e.g., full amortization, interest-only, negative amortization), income documentation type (e.g., full, low-doc, or no-doc), interest rate type (e.g., fixed or adjustable), prepayment penalty, loan purpose (e.g., purchase or refinance), reported occupancy status (e.g., owner-occupied, investment, or second-home), delinquency method (i.e., MBA or OTS), and delinquency status (e.g., current, 30, 60, 90

days, etc.). The definitions of variables constructed from this information are presented in Table 1.

2.5 Descriptive Statistics

We study the effect of gambling preference on second-lien misrepresentation in a sample that includes loans without simultaneous second liens, loans with correctly reported simultaneous second liens, and loans with misreported simultaneous second liens. We select the sample period of 2005 to 2007, during which mortgage misrepresentation measures can be calculated using the databases and misrepresentations are not rare. Table 2 shows the summary statistics. For each continuous variable, we report the number of observations, mean, standard deviation, 25th percentile, 50th percentile, and 75th percentile. For dummy variables, we report the number of observations and mean. The continuous variables are winsorized at the 0.5 percent level to ensure our results are not driven by extreme values.

In our sample, we have 3,031,921 loans for which the borrower’s area CPRATIO is available⁹. Among these loans, about 7.15 percent have misreported second liens, while 13.91 percent have reported second liens, resulting in a total of 21.06 percent. This percentage is comparable to the one reported by Griffin and Maturana (2016) (10.2%), who used a different database and included all loans (honestly reported, fraudulently hidden, and truly without second liens). Figure 1 plots the nationwide distribution of the county-level proportion of second-lien misrepresentation in all loans, showing variations both across and within states.

Our primary independent variable of interest is local gambling preference (CPRATIO). While there are 3,090 available U.S. counties from 2005 to 2007, we only retain county-year level data where loan-level data is also available. For the 8,716 data points, the mean CPRATIO is 0.51 and the median is 0.19, indicating a large positive skewness¹⁰. Figure 2 plots the nationwide distribution of CPRATIO. Although high gambling preference areas

⁹We also exclude cases where the LTV is greater than 99% since our measure is calculated using LTV. An LTV greater than 99% is likely erroneous or highly unlikely to have a simultaneous second lien.

¹⁰The mean and median CPRATIO from 1980 to 2005 are 0.60 and 0.23, respectively, in Kumar et al. (2011), which is quite close to our results if we include all counties.

are mostly concentrated on the west coast, southwest, and east coast, within-state variations exist in many states, such as Texas, Wisconsin, and Indiana.

One concern with using local gambling preference is its potential correlation with other geographic variables. To assess this, we calculate correlations between CPRATIO and county-level characteristics such as religiosity, unemployment, and house price appreciation (HPA). None of the correlations between CPRATIO and these variables exceed 0.32, indicating that CPRATIO contains information not solely covered by other common geographic characteristics.¹¹ This suggests that our categorization of counties by CPRATIO is unlikely to be driven by any single alternative factor.

3 Gambling Preference and Second-lien Misrepresentation

In this section, we investigate the relationship between gambling preference and second-lien misrepresentation. When making decisions, individuals often exhibit various behavioral traits, with gambling preference being prevalent in financial markets. Misreporting a second lien can be seen as a gamble during our sample period, as borrowers could achieve significant gains by betting on the success of such misrepresentation. On one hand, housing prices increased rapidly in the early 2000s, making the financial return of owning a house very attractive (Bostic and Lee, 2008; Goodman and Mayer, 2018). On the other hand, homeownership became an integral part of the American dream (Clinton, 1995; Bush, 2003), making the purchase of a house more than just an investment. Moreover, within the framework of the fraud triangle theory, high gambling preference fosters misrepresentation by providing borrowers with a rationalization channel (Christensen et al., 2018). Consequently, for borrowers with high gambling preferences who need simultaneous second liens to finance housing, misreporting a second lien may seem like a favorable choice. Thus, we expect

¹¹See Appendix for the full results.

that borrowers in areas with high local gambling preferences are more likely to misrepresent material information in mortgage applications. We formalize this idea in Hypothesis 1:

Hypothesis 1 *Second-lien misrepresentation is more likely to occur in counties with higher levels of local gambling preference.*

To test our hypotheses, we estimate loan-level linear regressions of the following form:

$$Y_{it} = \alpha + \beta CPRATIO_{ct} + \gamma X_{it} + \delta_s + \eta_t + \lambda_o + \epsilon \quad (1)$$

where Y_i is an indicator for second-lien misrepresentation on loan i originated at time t ; $CPRATIO_{ct}$ is the county level measure of gambling preference at time t ; X_{it} includes loan i 's religiosity, geographic controls, and loan characteristics at time t ; δ_s is state fix effects; η_t is origination half-year fixed effect (Piskorski et al., 2015); λ_o is originator fixed effects¹²; ϵ is an error term. Moreover, since our primary independent variable of interest (CPRATIO) is measured at the county level, we cluster heteroskedasticity-robust standard errors by county. We gradually include control variables and fixed effects in the specifications, with standard errors in all specifications clustered at the county level. Additionally, we winsorize all continuous variables at the 0.5 percent level and then standardize them.

Table 3 presents the results of regressions¹³. Column (1) presents the basic model with only control of simultaneous second lien. Column (2) adds geographic controls only. Column (3) adds loan characteristics controls only. Column (4) adds all control variables. Column (5) to (7) adds originator fixed effects, state fixed effects, and half-year fixed effects gradually.

Our results reveal that counties with higher levels of gambling preference indeed tend to have more second-lien misrepresentation. A one standard deviation increase in CPRA-

¹²Although both originators and underwriters play important roles (Griffin and Maturana, 2016) and the related fixed effects are used in different literature, we control for originator fixed effects because originators are the agency that interact with the borrowers so that they may affect the decision of borrowers while underwriters do not have such influence in the borrowers decision-making process.

¹³Following Piskorski et al. (2015) and Yavas and Zhu (2024), we use OLS regressions in the main tests despite the binary nature of the dependent variable because it has an important advantage: allow the inclusion of multiple fixed effects. By contrast, probit or logit models generally yield inconsistent estimates when more than one fixed effect is included in large samples (?).

TIO in the whole sample leads to a 0.12 percent increase in the probability of second-lien misrepresentation without fixed effects and a 0.09 percent increase with state, half-year, and originator fixed effects¹⁴. The significance and economic magnitude also show that CPRATIO is a leading variable among all county-level variables¹⁵. It is also comparable to loan-level control variables, such as interest rate (-0.73 percent with all fixed effects) and FICO score (-0.62 percent with all fixed effects).

Simultaneous second lien is a key control in all specifications because, to misrepresent, one must first have a simultaneous second lien. Indeed, the coefficients of simultaneous second lien are significant in all columns, showing that about one-third of simultaneous second liens are unreported. Turning to geographic controls, we see that religiosity, income, and education are consistently negatively, though not always significantly, related to second-lien misrepresentation. In contrast, total population and the proportion of the elderly population are consistently positively, though not always significantly, related to this type of mortgage misrepresentation. The relationship of other geographic characteristics varies in different situations. Moreover, except for our variable of interest (CPRATIO), only total population remains robustly significant. The loan-level characteristics show that second-lien misrepresentation is associated with lower interest rates after controlling for simultaneous second liens, indicating potential gains from misrepresenting second liens. Low credit scores, low initial balances, and high LTV ratios are associated with unreported second liens, indicating that second-lien misrepresentation occurs more frequently in loans with these characteristics.

¹⁴The magnitude of the coefficient is small because the total misrepresentation rate is low (7.15 percent). The increase corresponds to a 1.67 percent rise compared to the total misrepresentation rate.

¹⁵The coefficient of CPRATIO is larger than that of REL, which has been proven to be an important factor in mortgage misrepresentation by Conklin et al. (2022), and even renders REL insignificant in columns (6) and (7).

4 Interpreting the Effect: Borrower Behavior or Lender Practice

Previous studies examining second-lien misreporting in privately securitized loans show that both originators and underwriters play significant roles in mortgage misrepresentation (Griffin and Maturana, 2016; Piskorski et al., 2015). They are likely aware of hidden second liens but still misreport them. Additionally, by comparing lenders' portfolio loans and privately securitized mortgages, Yavas and Zhu (2024) provide strong evidence that second-lien misrepresentation occurs in the early stages of intermediation by lenders rather than underwriters. Building on these findings, it is important to assess whether the observed relationship should be interpreted through borrower behavior or lender practice. In this section, we present evidence from both perspectives and show that the results are more consistent with an interpretation based on borrower behavior.

4.1 Evidence from Borrower Behavior

According to Christensen et al. (2018), the impact of gambling preference on misreporting should be more pronounced in situations where individuals face stronger incentives. Since our interpretation primarily emphasizes the borrower's perspective, we examine whether the effect is amplified when borrowers' incentives are greater, using subsample analysis.

Specifically, we partition the sample by the loan's property occupancy type (primary or non-primary) and by the loan's purpose type (purchase or refinance). A primary occupancy type indicates that the property is owner-occupied, which is tied to the borrower's need for stable housing and the utility derived from living in the home. In contrast, a non-primary occupancy type implies that the property is used as a second home or investment, suggesting that the borrower already owns a residence and is motivated by discretionary consumption or wealth accumulation. Because the incentive to misrepresent should be greater when stable housing is at stake, we expect the effect of gambling preference is more evident in the primary

occupancy subsample. Similarly, if the loan purpose is a purchase, the borrower does not yet own the property and must obtain it to begin consumption or investment. By contrast, if the purpose is refinance, the borrower already owns the property and seeks to improve financial conditions by lowering costs or accessing home equity. In this case, the incentive to misrepresent should be stronger in purchase loans, making them a setting where gambling preference is expected to play a larger role.

Table 4 reports the subsample results. Columns (1) and (2) present estimates for the primary and non-primary groups. Most observations in the overall sample fall into the primary category. Moreover, only the coefficient of CPRATIO in the primary subsample is significant, and its economic magnitude is substantially larger than that in the non-primary group. This finding suggests that borrowers with stronger gambling preferences are more likely to misrepresent second liens when the property is intended as a primary residence, consistent with our expectation that the relationship between gambling preference and misrepresentation is stronger in the primary subsample, where borrower incentives are greater.

Columns (3) and (4) present the results for the purchase and refinance subsamples. About 43 percent of the observations come from the purchase group. Once again, only the coefficient of CPRATIO in the purchase subsample is significant, and its economic magnitude is much larger than that in the refinance group. This indicates that borrowers with stronger gambling preferences tend to misrepresent second liens primarily when purchasing a house, reinforcing our expectation that the relationship between gambling preference and misrepresentation is stronger in the purchase subsample, where borrower incentives are greater.

Overall, the subsample analysis shows that second-lien misrepresentation rises more sharply with gambling preference when borrowers face stronger incentives. This evidence support our interpretation of the effect from borrower’s perspective.

4.2 Evidence from Lender Practice

To assess this issue from the perspective of lenders' practices, we employ the setting of ease of securitization around a FICO score of 620, as described by Keys et al. (2010). Due to the underwriting guidelines established by government-sponsored enterprises, Fannie Mae and Freddie Mac, low documentation loans made to borrowers with FICO scores of 620 or higher are much easier to securitize. Consequently, lenders have lax screening standards for such loans, and a regression discontinuity design could reveal an upward jump in the default rate for low documentation loans. Additionally, Griffin and Maturana (2016) find that the second-lien misrepresentation rate also jumps at this cutoff, implying that this type of misrepresentation is associated with lenders' incentives to securitize the loan.

Utilizing this shock to lenders, we first study the extent to which lenders facilitate second-lien misrepresentation by comparing loans with misreported simultaneous seconds to other types of loans. Specifically, we examine the jumps in the number of loans and the jumps in the default rate for all loans, loans with correctly reported simultaneous seconds, and loans with misreported simultaneous seconds. If lenders specifically facilitate second-lien misrepresentation, then the jump ratio in the number of loans with misreported simultaneous seconds should be greater than for other types of loans. Additionally, if such special facilitation of misrepresentation stems from lenders' screening efforts, the jumps in the default rate for loans with misreported simultaneous seconds should be significantly greater than those for loans with correctly reported simultaneous seconds. In contrast, if the jump ratio in the number of loans with misreported simultaneous seconds is similar to or smaller than for other types of loans, lenders do not specifically facilitate second-lien misrepresentation but increase it with the increase of other loans. Finally, if the jumps in the default rate for loans with misreported simultaneous seconds are no larger than those for loans with correctly reported seconds, this would imply that lenders' screening effort does not play a distinct role in facilitating misrepresentation.

Building on this general picture, we next investigate whether the effect of gambling pref-

erence on second-lien misrepresentation aligns with lender practices. Since the variation in second-lien misrepresentation related to gambling preference could come from either lenders or borrowers, and the shock we employed is only for lenders, if the increase in second-lien misrepresentation in high gambling preference areas is smaller, it indicates that lenders play a smaller role in increasing misrepresentation in high gambling preference areas. This implies that the positive relationship between gambling preference and second-lien misrepresentation is mainly attributed to borrowers. Specifically, we examine the jumps in the number of loans and the jumps in the default rate to understand how lender's roles differ across areas. If the effect of local gambling preference on second-lien misrepresentation is in line with lenders' practice (i.e., lenders make second-lien misrepresentation more prevalent in high gambling preference areas), this shock to lenders would lead to different outcomes between high and low gambling preference areas, favoring high gambling preference areas (i.e., a greater increase in the number of loans with misrepresentation and a larger increase in the default rate). In contrast, a smaller increase in the number of misreported loans and the default rates of misreported loans in high gambling preference areas indicates that lenders did not facilitate more misrepresentation in such areas.

We take RDD approach using local linear regressions. The credit scores are normalized as follows:

$$C = Credit\ Score - Threshold \quad (2)$$

where *Threshold* is 620 for low documentation loans. Then, we define D to distinguish credit scores that are over or below the threshold as follows:

$$D = \begin{cases} 1, & \text{if } C \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

To differentiate the effect between high and low local gambling preference areas, we define high local gambling preference areas:

$$G = \begin{cases} 1, & \text{if } CPRATIO \geq 1.2921 \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

We use 1.2921 as the cutoff for two reasons¹⁶. First, only 10 percent of county-years in our sample have a CPRATIO greater than or equal to 1.2921, indicating that areas with such a CPRATIO indeed have a high local gambling preference. Second, about half of all loans in our sample are originated in these areas, making our findings about high and low gambling preference areas comparable.

To apply the RDD approach using local linear regressions, we choose a bandwidth of 10, which corresponds to a FICO score range of 610 to 630. We selected this window because it ensures that no other jumps are found in the literature¹⁷ and it is close to the optimal bandwidth generated by a data-driven bandwidth selection method¹⁸ (Calonico et al., 2020). We use the fixed bandwidth rather than the optimal bandwidth so that the RDD results in different cases are comparable¹⁹. For the default rate and number of loans in high and low gambling preference areas, we estimate the four cases separately using the following specification:

$$Y = \alpha + \beta D + \gamma C + \delta D \times C + \epsilon \quad (5)$$

where Y is the default dummy variable for loan i originated at time t or the number of loans at each FICO score. In our baseline model, we use uniform kernel for estimation.

Finally, to calculate the difference in the discontinuities (diff-in-disc) between high and low gambling preference areas, we follow Dickert-Conlin and Elder (2010) and Grembi et al. (2016) to estimate the following model:

¹⁶We also tried other reasonable cutoffs, such as 1, and the results are robust.

¹⁷For example, FICO scores of 600 and 660 could be other potential cutoffs for jumps.

¹⁸The optimal bandwidth for the number of all loans is 10.50.

¹⁹Using the optimal bandwidth does not affect our findings.

$$Y = \beta_0 + \beta_1 D + \beta_2 C + \beta_3 D \times C + \beta_4 G + \beta_5 G \times C + \beta_6 G \times D + \beta_7 G \times C \times D + \epsilon \quad (6)$$

where β_6 is the parameter that measures the difference in the discontinuity in Y_{it} at credit score cutoff for high versus low gambling preference areas. When we apply this equation to calculate the difference in jump ratios between high and low gambling preference areas, we first rescale the data to obtain the t-statistics. We divide the data in high gambling preference areas by the estimated value at FICO 620⁻ in those areas. Similarly, we divide the data in low gambling preference areas by their corresponding estimated value at FICO 620⁻. After this rescaling, the discontinuity estimated in high or low gambling preference areas can be directly interpreted as a multiple of its corresponding estimated value at FICO 620⁻. The difference between the two discontinuities represents the difference in jump ratios between high and low gambling preference areas.

We first look at the jumps in the number of loans and the jumps in the default rate in all areas, examining the pattern of second-lien misrepresentation given a shock to lenders. Panel A of Table 5 presents the results for jumps in the total number of loans, loans with correctly reported simultaneous seconds, and loans with misreported simultaneous seconds. Overall, the total number of low-documentation loans doubles from 620⁻ to 620⁺, consistent with the findings in Keys et al. (2010). Loans with simultaneous seconds increase more than those without, whether correctly reported or misreported. This shows that the ease of securitization is indeed applicable to such loans. Moreover, as shown in Panel A of Figure 3, the jump ratio of misreported simultaneous seconds is less than the jump ratio of correctly reported simultaneous seconds, indicating that lenders do not facilitate second-lien misrepresentation more than they do correctly reported seconds.

Panel B of Figure 3 compares the jump in default rates between correctly reported and misreported seconds. The figure shows that the increase in default rates is less pronounced

for loans with misreported seconds than for those with correctly reported seconds. Panel B of Table 5 reports regression estimates with control variables.²⁰ Consistent with previous literature, there is a jump of 0.037 in the default rate for all loans, indicating overall reduced screening for 620⁺ loans. This lax screening is also observed in loans with correctly reported simultaneous seconds, with a statistically significant jump of 0.044. For the loans with misreported simultaneous seconds, the jump in the default rate is similar in magnitude but statistically insignificant, suggesting that the lax screening effect is not as pronounced as for correctly reported seconds. These findings illustrate that lenders facilitate loans with simultaneous second liens, whether correctly reported or misreported, and that the lax screening effect is no larger for misreported seconds than for correctly reported seconds.

After looking at the jumps in all areas, we further explore the patterns between high and low gambling preference areas to determine if the effect aligns with lender practices. Table 6 shows the results for the number of loans in Panel A and the results for default rates in Panel B. For the number of loans, the magnitude of increases is similar between high and low gambling preference areas for all loans and loans with correctly reported simultaneous seconds. This indicates no systematic difference between high and low gambling preference areas in ease of securitization. In contrast, as shown in Panel A of Figure 4, the magnitude of the increase in the number of loans with misreported seconds in high gambling preference areas is much smaller than in low gambling preference areas. Given that the ease of securitization is a shock to lenders (Keys et al., 2010) and lenders facilitate misrepresentation with the objective of securitization (Griffin and Maturana, 2016), this result suggests that lenders do not play a larger role in increasing second-lien misrepresentation in high gambling preference areas²¹.

²⁰Figures 3 and 4 present unconditional results with conventional t-statistics to illustrate the pattern of jumps in default rates relative to their levels. Tables 5 and 6 present conditional results with bias-corrected t-statistics for more precise estimation.

²¹Since borrowers play a larger role in increasing second-lien misrepresentation in high gambling preference areas, lenders' effect in increasing the number is more limited, making the magnitude of the jump smaller. In contrast, since borrowers play a smaller role in increasing second-lien misrepresentation in low gambling preference areas, lenders' effect in increasing the number is less limited, making the magnitude of the jump bigger

For default rates, the jumps between high and low gambling preference in all types of loans do not show significant differences. Additionally, for loans with misrepresented simultaneous seconds, the jump in high gambling preference areas is smaller than in low gambling preference areas, as shown in Panel B of Figure 4, although the difference is not significant. This suggests that lenders in high gambling preference areas are not more likely to facilitate misrepresentation by having laxer screening in such areas. Therefore, taken together with the evidence on the number of loans, these findings suggest that lenders in high gambling preference areas are not more inclined to increase such misrepresentation than lenders in low gambling preference areas. In other words, the evidence does not support the interpretation that the effect operates through lenders’ practices.

5 Outcomes of Gambling Preference-Related Misrepresentation

In the previous section, we found a significant correlation between gambling preference and mortgage misrepresentation. Our next step is to investigate the outcomes of gambling preference-related mortgage misrepresentation. Prior studies show that mortgage misrepresentation generally leads to worse loan performance (Piskorski et al., 2015; Griffin and Maturana, 2016), but it is unclear whether the misrepresentation associated with gambling preference has systematically different consequences. Since gambling is typically linked to adverse financial outcomes, we conjecture that gambling preference-related misrepresentation exacerbates loan performance problems. We formalize this inference in Hypothesis 2:

Hypothesis 2 *The adverse performance effects of second-lien misrepresentation are stronger in counties with higher levels of gambling preference.*

To test the hypotheses, we include an interaction term between mortgage misrepresentation measure and gambling preference measure in the form of equation 1:

$$\begin{aligned}
Y_{it} = & \alpha + \beta \textit{Misreported Second}_{it} + \gamma \textit{CPRATIO}_{ct} \\
& + \kappa \textit{Misreported Second}_{it} \times \textit{CPRATIO}_{ct} + \phi X_{it} + \delta_s + \eta_t + \lambda_o + \epsilon
\end{aligned} \tag{7}$$

where Y_i is the default variable on loan i originated at time t , which equals one if the loan becomes 90 days or more delinquent using MBA method and zero otherwise. For independent variables, $\textit{Misreported Second}_{it}$ is the second-lien misrepresentation indicator on loan i originated at time t ; $\textit{CPRATIO}_{ct}$ is the county level measure of gambling preference at time t ; $\textit{Misreported Second}_{it} \times \textit{CPRATIO}_{ct}$ is the interaction term between mortgage misrepresentation measure and gambling preference measure; X_{it} includes loan i 's correctly reported simultaneous second, geographic controls, and loan characteristics at time t ; δ_s is state fix effects; η_t is origination time fixed effects for half year; λ_o is originator fixed effects; ϵ is an error term. Moreover, since our primary independent variable of interest (CPRA-TIO) is measured at the county level, we also cluster heteroskedasticity-robust standard errors by county. All continuous variables are winsorized at the 0.5 percent level and then standardized. We include control variables, fixed effects, and standard error clustering in all specifications, but we gradually include the mortgage misrepresentation term, gambling preference term, and the interaction term.

The results are reported in Table 7. Column (1) includes the misrepresentation indicator only and confirms that loans with misreported second liens default at significantly higher rates than other loans. Column (2) adds the gambling preference measure as a baseline control. Column (3) introduces the interaction term between misrepresentation and gambling preference, which is the main focus of our analysis.

The coefficient on second-lien misrepresentation is significantly positive across all specifications, indicating that loans with misreported seconds default at higher rates. The interaction between misrepresentation and gambling preference is also positive and significant, showing that the default penalty associated with misrepresentation is larger in counties with

higher gambling preference. This suggests that gambling preference amplifies the negative performance consequences of misrepresentation.

To further illustrate the economic magnitude of our estimates, we do a back-of-envelope calculation. A one-standard-deviation increase in CPRATIO raises the default probability of misreported loans by about 1.4 percentage points. Given 216,736 such loans nationally in the sample, this translates into roughly 3,034 additional defaults. At an average first-lien balance of approximately \$330,000, this corresponds to about \$0.8 billion in first-lien balance for private-label RMBS during the crisis period. This magnitude illustrates that the interaction of gambling preference and mortgage misrepresentation is not only statistically significant but also economically meaningful.

6 Robustness

6.1 Alternative Gambling Preference Measures

While CPRATIO serves as our primary measure of gambling preference, we also employ alternative measures to address concerns that CPRATIO may reflect broader religious differences rather than gambling-specific tendencies. Following studies such as Markham et al. (2023), which use the density of gambling outlets as a measure of gambling exposure, we incorporate employment and the number of gambling establishments as alternative measures that reflect gambling preference through local supply. Specifically, we utilize per capita employment in the gambling industry and per capita number of gambling establishments at the core-based statistical area (CBSA) level. These alternatives help isolate local gambling behavior more directly. We regress the second-lien misrepresentation dummy variable on these alternative measures, alongside geographic controls at the CBSA level, loan-level controls, and fixed effects (i.e., state, lender, and half-year). Additionally, we include controls for employment and establishment counts in casino hotels to account for tourist-driven gambling demand²².

²²In the CBP database, casino hotels are categorized separately from the gambling industry. Because casino hotel employees also provide hotel-related services beyond gambling, it is necessary to control for

We also include nonemployer gambling establishments to capture complementary gambling demand that may be excluded by the primary measures due to database limitations. See Appendix for detailed descriptions of the data and empirical tests.

Table 8 presents the results of the primary regressions. Columns (1) and (2) report specifications using per capita employment in the gambling industry and per capita number of gambling establishments, respectively. Columns (3) and (4) include both employment and establishment metrics for the gambling industry and casino hotels, as well as nonemployer gambling establishments. The regression results indicate that both employment and establishment metrics for the gambling industry are significantly positive, suggesting a strong link between local gambling preference and the likelihood of second-lien misrepresentation. By contrast, casino hotel variables, which reflect a mix of local and tourist demand, yield coefficients that are generally insignificant. Although nonemployer gambling businesses primarily reflect local demand, they tend to offer limited gambling access, and their associated coefficients are likewise insignificant. Overall, we find that per capita employment and establishment measures in the gambling industry, which are used as alternative proxies for local gambling preference, support the hypothesis that second-lien misrepresentation is more prevalent in regions with elevated gambling preference.

6.2 Consistency with Credit Expansion and Belief Distortion

While prior literature emphasizes credit expansion and belief distortion as explanations for the rapid rise in household leverage before the 2008 financial crisis, this paper adds the perspective of borrower behavioral traits. To assess whether our findings are consistent with these existing views, we examine the relationship between gambling preference and second-lien misrepresentation in related subsamples. Moreover, if we take the credit expansion and belief distortion views as given, the subsample analysis further supports the borrower-side interpretation of gambling preference, since the partitions also reflect variation in borrowers' casino hotels separately.

incentives to misreport.

Specifically, we partition the sample by area income (high or low income), credit score (high or low FICO), and past house price appreciation (high or low HPA). The high- and low-income subsamples are defined by comparing local household median income with the sample median at the ZIP code level. The high- and low-credit-score subsamples are divided using a FICO cutoff of 670²³. The high- and low-HPA subsamples are defined based on the state-year median HPA, calculated from all available ZIP code-level data.

Area income and credit score are closely tied to the credit expansion argument, which centers on whether the expansion of mortgage credit leading up to the crisis was widespread across the population. In our setting, gambling preference should affect all groups, as all contributed to the crisis. However, low-income and low-credit-score borrowers are expected to show stronger effects because they have greater incentives to misrepresent, stemming from tighter financial constraints (low income) or more limited credit access (low FICO). As shown in Table 9, columns (1)–(4), the economic magnitude of CPRATIO is large²⁴, and the effects are larger and more significant in the low-income and low-FICO subsamples. This pattern is consistent with the credit expansion view and also reinforces the interpretation of borrower gambling preference.

Past house price appreciation is closely related to the belief distortion channel, which emphasizes that households during this period held overly optimistic expectations of future price growth. In our context, households in areas with higher past appreciation are more likely to extrapolate those increases into the future (Shiller, 2007), creating stronger incentives to misrepresent due to higher expected returns. As shown in Table 9, columns (5)–(6), the coefficient of CPRATIO is larger in magnitude and more statistically significant in high-HPA areas. This evidence indicates that our findings are also consistent with the belief distortion channel, while continuing to support the interpretation of borrower gambling

²³670 is the boundary between fair and good levels as evaluated by the institution. We also tried the median of the sample, 682, which yielded similar results.

²⁴Recall that in the baseline results, the coefficient of CPRATIO is about 0.0009.

preference.

6.3 Subsample Evidence on Outcomes of Gambling Preference

While Section 5 shows that gambling in mortgage applications generally leads to worse outcomes, we now examine whether this effect varies across subsamples.

Table 10 reports results for subsamples related to the credit expansion and belief distortion arguments. The interaction terms, which measure how default rates change with gambling preference conditional on the occurrence of second-lien misrepresentation, are economically meaningful and statistically significant across all subsamples. These findings indicate that regardless of income class or the extent of belief distortion, when misrepresentation is associated with high gambling preference, loan outcomes are worse.

6.4 Alternative Measures of Default

Borrowers can default on their mortgages for various reasons over different time horizons. Our measure of default uses the MBA method of delinquency for 90 days or more and is restricted to the first three years after origination. By adjusting the definition to be more restricted (fewer default cases) or less restricted (more default cases), we can investigate whether the effect found in section 5 is generally applicable. Therefore, we consider the following default measures: 90 days or more delinquency using the MBA method in the first two years after origination (more restricted), 60 days or more delinquency using the MBA method in the first three years after origination (less restricted), and bankruptcy/foreclosure/REO in the first three years after origination (more restricted).

Table 11 presents the results of different default measures. We also include the results of the original measure in column (1). The significance of second-lien misrepresentation and its interaction with CPRATIO are robust. Additionally, when the default measure is less restricted, the effect of mortgage misrepresentation is greater. The effect of gambling preference-associated mortgage misrepresentation is also greater when the default measure

is less restricted.

6.5 Nonlinear Model

The results in the previous sections are estimated using a linear probability model (OLS). To ensure our findings are not driven by this modeling choice, we also use a nonlinear specification (probit) for inference. We present the marginal effects of CPRATIO in Table 12. Since we control for simultaneous second liens in the regression, the probit model drops all observations without simultaneous seconds due to perfect failure prediction of misrepresentation. The marginal effects of CPRATIO remain significant.

6.6 Causal Forest and Causality

To enhance the causal inference between gambling preference and mortgage misrepresentation and to explore the heterogeneous treatment effects of mortgage misrepresentation on mortgage default conditioned on gambling preference, we employ the causal forest approach proposed by Wager and Athey (2018). This method uses an augmented inverse propensity-weighted estimator with the random forest method in machine learning, incorporating an honesty condition. It provides double robustness (compared to propensity score matching) and high efficiency for high-dimensional models (compared to nearest-neighbor matching) in observational settings²⁵. This approach has also been used in finance, such as in Rampini and Viswanathan (2022) for secured debt and Gulen et al. (2021) for corporate finance, and has shown better performance than traditional causal inference designs (Gulen et al., 2021) in terms of accuracy.

We first use this approach to investigate the causality between gambling preference and mortgage misrepresentation. In the causal forest approach, we set mortgage misrepresentation as the outcome and CPRATIO as the treatment. We use all control variables from

²⁵See Wager and Athey (2018) for statistical illustration. See Athey and Wager (2019) for an example application.

Table 3 as matching variables to help grow trees and forests²⁶. All continuous variables, including CPRATIO, are winsorized at the 0.5 percent level and then standardized. Since fixed effects are not applicable in this approach, we create a variable for half-year periods, starting from the first half of 2005 as 1, to account for potential time effects. Similar to Athey and Wager (2019), who cluster observations by school ID, we cluster observations by state but give each unit the same weight so that larger clusters receive more weight²⁷. By using such cross-fitting, our results do not solely come from any single state, but states with a greater number of observations do have greater weights. To balance computational capacity and the accuracy of confidence intervals, we grow 2000 trees for the forest, which is also the default setting in Athey and Wager (2019). For the honesty property, we use the default splitting fraction: for each sample, we use 50 percent of the data for splitting and the remaining data for estimation. Table 13 shows the results using the causal forest. The coefficients in the regressions of second-lien misrepresentation remain significant, and the magnitude is similar to those estimated by OLS.

Second, we study the heterogeneous treatment effects of mortgage misrepresentation on default conditioned on different levels of gambling preference. We set default as the outcome and mortgage misrepresentation as the treatment. Except for adding CPRATIO to the matching variable matrix, the other matching variables and clustering variables are the same as those in the above causal forest regressions. For the same considerations, we also grow 2000 trees for the forest and use 50 percent of the data in each sample for splits. After estimating the causal forest, the average treatment effects of mortgage misrepresentation on default are calculated for each quarter of the sample sorted by CPRATIO. Figure 5 shows the results for second-lien misrepresentation. The treatment effects are generally larger when CPRATIO is greater, which is consistent with the results from the OLS model.

²⁶We require observations to be non-missing for all variables.

²⁷Due to the limitation of the grf package in R, which requires the matching variables to be numerical, we do not create a numerical variable for originators to avoid falsely treating closer values as shorter distances. We also do not cluster observations by originator because the large number of originators would excessively increase the computational burden.

6.7 Diff-in-disc concerns

To validate our application of the diff-in-disc approach in comparing loan volumes and default rates between high and low gambling preference areas, we address several methodological concerns. First, we examine and test the three core assumptions outlined by Grembi et al. (2016) regarding the implementation of the diff-in-disc framework. Second, we assess the sensitivity of our results to key modeling choices, including the inclusion of control variables, bandwidth selection, and estimation kernel specification. Our findings remain robust across these variations. Third, we perform a placebo test to confirm that the observed effects are not driven by random variation. For full details of these robustness checks and supporting evidence, see Appendix.

7 Conclusion

Many years after the 2008 financial crisis, the housing market continues to exhibit cyclical dynamics and remains a significant driver of global economic conditions. Although considerable progress has been made in understanding the origins of the crisis, debate persists over which segments of society played the central role. By shifting the focus from income classes to behavioral traits, this paper highlights a trait-based behavioral channel that complements the credit expansion and expectations views of the crisis.

Using a religion-based proxy for local gambling preferences, we show that gambling preference, a persistent behavioral trait, is associated with higher rates of second-lien misrepresentation across borrower classes, ultimately leading to poorer loan performance. Our findings suggest that behavioral predispositions shape borrower actions in ways that contribute to systemic risk. Recognizing these traits is essential for designing policies that address the behavioral foundations of financial instability.

High household leverage was an important feature of the housing boom preceding the financial crisis. A particularly striking form of excessive leverage taking was the misrepresen-

tation of second liens, which allowed borrowers to overstate their borrowing capacity and obtain mortgages beyond sustainable limits. This paper examines how borrower heterogeneity shaped the decision to take on such high leverage.

Using a religion-based proxy for gambling attitudes, we find that second-lien misrepresentation is more prevalent in gambling-prone counties. The relationship is especially strong when borrower incentives to misreport are high. In contrast, evidence from the FICO 620 securitization cutoff provides little support for lender-based explanations. We also show that the negative performance consequences of misrepresentation are amplified in high-gambling-preference areas. Taken together, the results suggest that borrower behavioral traits, specifically gambling preference, played a significant role in fueling excessive leverage before the crisis. Recognizing the influence of such traits is important for understanding the fragility of mortgage markets and the behavioral foundations of financial instability.

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Table 1
Variable Definitions

Variable name	Definition
Mortgage Reporting	
Misreported second	Indicator that equals one if the borrower has simultaneous second-liens and misreported the second-liens on the loan application
Reported second	Indicator that equals one if the borrower has simultaneous second-liens and correctly reported the second-liens on the loan application
Simul. second	Indicator that equals one if the borrower have simultaneous second-liens regardless the reporting status on the loan application
Gambling preference and religiosity	
CPRATIO	Ratio of the county’s Catholic residents to Protestant residents
REL	Proportion of the county population that are religious adherents
Geographic controls	
HPA	Zip code house price appreciation in the two years prior to loan origination year. County level data used if zip code index is not available
Unemployment	Unemployment rate of the county
Education	Proportion of the county population over age 25 that has completed a bachelor’s degree or higher
Married	Proportion of the county population over age 15 that is married
Income	Natural logarithm of the zip code level median household income
Urban	Proportion of the county population that lives in urban area
Total population	Natural logarithm of the county total population
Over65	Proportion of the county population over age 65
Male-female ratio	Ratio of the county’s male residents to female residents
Minority	Proportion of the county non-white residents
Loan characteristics	
Interest rate	Loan interest rate at origination
FICO	Natural logarithm of borrower’s FICO credit score at loan origination
Balance	Natural logarithm of the initial loan balance
LTV	Loan-to-value ratio at loan origination
ARM	Indicator that equals one if the loan is an adjustable rate mortgage
Option ARM	Indicator that equals one if the loan has an ARM convertibility clause
Negative amortization	Indicator that equals one if the loan allows negative amortization
Low or no doc.	Indicator that equals one if the loan is originated with no or limited documentation
Prepayment penalty	Indicator that equals one if the loan would be assessed a penalty on any early voluntary prepayment
Cash-out	Indicator that equals one if the loan purpose is cash out refinancing
No-cash-out	Indicator that equals one if the loan purpose is no cash out refinancing
Investment	Indicator that equals one if the loan occupancy status is investment
Second-home	Indicator that equals one if the loan occupancy status is second-home
Default	Indicator that equals one if the loan becomes 90 days or more delinquent using MBA method in the first three years after origination

The table reports the variable definitions used in the empirical analysis part.

Table 2
Descriptive Statistics

	N	Mean	SD	P10	P25	P50	P75	P90
Simultaneous second liens (loan level)								
Misreported second (%)	3,031,921	7.15	25.76					
Reported second (%)	3,031,921	13.91	34.61					
Simul. second (%)	3,031,921	21.06	40.77					
Loan characteristis (loan level)								
Interest rate (%)	3,026,958	6.41	2.12	2.75	5.88	6.50	7.55	8.75
FICO	2,993,630	681.78	72.60	578.00	632.00	688.00	739.00	776.00
Balance (ln)	3,031,921	12.40	0.73	11.44	11.87	12.40	12.98	13.31
LTV (%)	3,031,921	74.82	13.34	56.65	70.00	80.00	80.00	90.00
ARM (%)	3,031,921	58.87	49.21					
Option ARM (%)	3,031,921	0.24	4.84					
Negative amortization (%)	3,031,921	13.58	34.26					
Low or no doc. (%)	3,031,921	65.42	47.56					
Prepayment penalty (%)	3,031,921	40.99	49.18					
Cash-out (%)	3,031,921	40.66	49.12					
No-cash-out (%)	3,031,921	14.73	35.44					
Investment (%)	3,031,921	10.27	30.36					
Second-home (%)	3,031,921	3.55	18.49					
Default (%)	3,014,278	23.51	42.40					
Geographic characteristics (county-year level)								
CPRATIO	8,716	0.51	0.88	0.01	0.04	0.19	0.54	1.29
REL (%)	8,716	59.59	17.64	36.62	45.90	58.51	72.42	89.21
Geographic characteristics (county-month level)								
Unemployment (%)	71,156	5.14	1.73	3.20	3.90	4.90	6.00	7.50
Geographic characteristics (county level)								
Education (%)	3,035	16.64	7.51	9.70	11.20	14.40	19.30	26.60
Married (%)	3,035	60.39	5.10	53.90	57.90	61.30	63.90	66.00
Urban (%)	3,035	40.62	30.54	0.00	12.93	40.32	64.65	84.67
Total population (ln)	3,035	10.49	1.11	9.52	9.52	10.15	11.05	12.09
Over65 (%)	3,035	14.71	4.05	9.90	12.10	14.40	17.00	20.00
Male-female ratio	3,035	0.98	0.06	0.92	0.94	0.97	1.00	1.05
Minority (%)	3,035	15.32	15.80	2.02	3.34	8.71	22.96	37.90
Geographic characteristics (zipcode-year level)								
HPA	76,535	18.45	14.28	4.23	7.55	13.93	26.56	39.99
Geographic characteristics (zipcode level)								
Income (ln)	26,517	10.57	0.34	10.17	10.34	10.53	10.77	11.02

The table presents descriptive statistics for the variables used in our study, covering the sample period from 2005 to 2007. For all continuous variables, we report the number of observations, mean, standard deviation, 25th percentile, 50th percentile, and 75th percentile. For all dummy variables, we report the number of observations, mean, and standard deviation. The variables are winsorized at the 0.5 percent level. (ln) indicates that the value of the variable is the natural logarithm of the original value. (%) indicates that the value of the variable is expressed as a percentage.

Table 3
Gambling Preference and Second-lien Misrepresentation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
CPRATIO	0.0012*** (0.0005)	0.0023*** (0.0005)	0.0023*** (0.0003)	0.0022*** (0.0005)	0.0016*** (0.0003)	0.0012** (0.0006)	0.0009** (0.0004)
REL		-0.0019*** (0.0006)		-0.0015*** (0.0005)	-0.0012*** (0.0004)	-0.0004 (0.0004)	-0.0001 (0.0003)
Unemployment		0.0041*** (0.0007)		0.0035*** (0.0007)	0.0018*** (0.0005)	0.0051*** (0.0009)	-0.0006 (0.0005)
HPA		-0.0071*** (0.0010)		-0.0050*** (0.0008)	-0.0008 (0.0006)	-0.0034*** (0.0005)	0.0017*** (0.0005)
Education		0.0004 (0.0006)		0.0011** (0.0005)	0.0003 (0.0004)	0.0015*** (0.0005)	0.0002 (0.0003)
Married		0.0006 (0.0006)		-0.0000 (0.0006)	0.0004 (0.0004)	0.0004 (0.0005)	-0.0000 (0.0003)
Income		-0.0003 (0.0003)		-0.0002 (0.0003)	-0.0004* (0.0002)	-0.0004 (0.0003)	-0.0004* (0.0002)
Urban		0.0005 (0.0005)		-0.0000 (0.0004)	0.0000 (0.0003)	-0.0009** (0.0004)	-0.0003 (0.0003)
Total population		0.0026*** (0.0009)		0.0028*** (0.0008)	0.0025*** (0.0006)	0.0039*** (0.0005)	0.0009** (0.0004)
Over65		0.0035*** (0.0005)		0.0036*** (0.0004)	0.0018*** (0.0003)	0.0001 (0.0004)	0.0005* (0.0003)
Male-female ratio		0.0003 (0.0006)		0.0005 (0.0005)	0.0011*** (0.0004)	-0.0000 (0.0004)	-0.0001 (0.0003)
Minority		-0.0006 (0.0008)		-0.0006 (0.0007)	0.0001 (0.0005)	-0.0020*** (0.0007)	0.0005 (0.0004)
Interest rate			-0.0127*** (0.0005)	-0.0119*** (0.0004)	-0.0121*** (0.0006)	-0.0113*** (0.0005)	-0.0073*** (0.0004)
FICO			-0.0092*** (0.0004)	-0.0093*** (0.0004)	-0.0071*** (0.0004)	-0.0072*** (0.0004)	-0.0062*** (0.0004)
Balance			-0.0005 (0.0004)	0.0011*** (0.0004)	0.0030*** (0.0002)	0.0027*** (0.0003)	0.0048*** (0.0002)
LTV			0.0040*** (0.0004)	0.0036*** (0.0004)	0.0056*** (0.0003)	0.0058*** (0.0003)	0.0056*** (0.0003)
ARM			0.0022*** (0.0008)	0.0017** (0.0009)	0.0121*** (0.0008)	0.0114*** (0.0008)	0.0088*** (0.0006)
Negative amortization			-0.0301*** (0.0013)	-0.0281*** (0.0013)	-0.0151*** (0.0020)	-0.0130*** (0.0018)	-0.0043*** (0.0012)
Option ARM			-0.0486*** (0.0031)	-0.0492*** (0.0032)	-0.0433*** (0.0046)	-0.0427*** (0.0046)	-0.0227*** (0.0049)
Prepayment penalty			-0.0093*** (0.0008)	-0.0086*** (0.0009)	-0.0096*** (0.0008)	-0.0106*** (0.0008)	-0.0081*** (0.0008)
Low or no doc.			-0.0027*** (0.0006)	-0.0023*** (0.0005)	-0.0069*** (0.0005)	-0.0072*** (0.0005)	-0.0080*** (0.0004)
Cash-out			0.0178*** (0.0006)	0.0182*** (0.0006)	0.0182*** (0.0005)	0.0180*** (0.0005)	0.0182*** (0.0006)
No-cash-out			0.0151*** (0.0010)	0.0144*** (0.0010)	0.0151*** (0.0007)	0.0153*** (0.0007)	0.0158*** (0.0007)
Investment			0.0232*** (0.0010)	0.0234*** (0.0011)	0.0256*** (0.0009)	0.0252*** (0.0009)	0.0258*** (0.0009)
Second-home			0.0108*** (0.0012)	0.0106*** (0.0011)	0.0138*** (0.0010)	0.0138*** (0.0010)	0.0157*** (0.0010)
Simul. second	0.3396*** (0.0020)	0.3408*** (0.0019)	0.3370*** (0.0019)	0.3388*** (0.0018)	0.3098*** (0.0019)	0.3105*** (0.0019)	0.3131*** (0.0019)
Originator FE	N	N	N	N	Y	Y	Y
State FE	N	N	N	N	N	Y	Y
Half-year FE	N	N	N	N	N	N	Y
Observations	3,031,921	2,907,303	2,991,961	2,869,014	2,384,686	2,384,686	2,384,686
Adj. R ²	0.289	0.291	0.282	0.284	0.313	0.313	0.318

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. Specific fixed effects are used if indicated by Y and not used if indicated by N. Variables are defined in Table 1. All continuous variables are winsorized at the 0.5 percent level and then standardized. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 4
Gambling Preference and Second-lien Misrepresentation - Incentive

	(1) Primary	(2) Non-primary	(3) Purchase	(4) Refinance
CPRATIO	0.0009** (0.0004)	0.0005 (0.0007)	0.0014** (0.0007)	0.0004 (0.0003)
REL	-0.0001 (0.0003)	-0.0003 (0.0006)	-0.0004 (0.0005)	0.0001 (0.0002)
Unemployment	-0.0006 (0.0005)	-0.0008 (0.0006)	-0.0007 (0.0009)	-0.0000 (0.0003)
HPA	0.0017*** (0.0006)	-0.0000 (0.0007)	0.0021** (0.0010)	0.0019*** (0.0004)
Education	0.0003 (0.0003)	-0.0013** (0.0005)	0.0002 (0.0006)	0.0002 (0.0002)
Married	0.0001 (0.0004)	-0.0007 (0.0005)	0.0002 (0.0006)	-0.0002 (0.0002)
Income	-0.0010** (0.0003)	0.0017** (0.0004)	-0.0003 (0.0004)	-0.0004** (0.0002)
Urban	-0.0004 (0.0003)	0.0013** (0.0006)	-0.0000 (0.0006)	-0.0004** (0.0002)
Total population	0.0010** (0.0004)	-0.0008 (0.0008)	0.0018** (0.0007)	0.0007** (0.0003)
Over65	0.0005 (0.0003)	0.0003 (0.0005)	-0.0001 (0.0004)	0.0004** (0.0002)
Male-female ratio	-0.0001 (0.0003)	0.0007 (0.0005)	-0.0001 (0.0005)	0.0003 (0.0002)
Minority	0.0006 (0.0005)	-0.0006 (0.0007)	0.0007 (0.0008)	0.0002 (0.0003)
Interest rate	-0.0088*** (0.0004)	-0.0000 (0.0007)	-0.0157*** (0.0008)	-0.0015*** (0.0003)
FICO	-0.0063*** (0.0004)	-0.0043*** (0.0004)	-0.0124*** (0.0007)	-0.0023*** (0.0003)
Balance	0.0052*** (0.0003)	0.0019*** (0.0004)	0.0061*** (0.0004)	0.0016*** (0.0002)
LTV	0.0050*** (0.0003)	0.0090*** (0.0004)	0.0172*** (0.0007)	0.0010*** (0.0002)
ARM	0.0116*** (0.0007)	-0.0153*** (0.0011)	0.0191*** (0.0012)	0.0009*** (0.0003)
Negative amortization	-0.0094*** (0.0014)	0.0315*** (0.0019)	-0.0247*** (0.0026)	0.0051*** (0.0007)
Option ARM	-0.0253*** (0.0055)	-0.0107** (0.0053)	-0.0562*** (0.0096)	-0.0061** (0.0029)
Prepayment penalty	-0.0076*** (0.0010)	-0.0088*** (0.0009)	-0.0102*** (0.0016)	-0.0047*** (0.0005)
Low or no doc.	-0.0094*** (0.0005)	-0.0089*** (0.0009)	-0.0084*** (0.0011)	-0.0040*** (0.0003)
Cash-out	0.0148*** (0.0006)	0.0308*** (0.0010)		
No-cash-out	0.0134*** (0.0007)	0.0264*** (0.0011)		
Investment			0.0256*** (0.0012)	0.0192*** (0.0007)
Second-home			0.0169*** (0.0013)	0.0102*** (0.0008)
Simul. second	0.3032*** (0.0020)	0.4117*** (0.0062)	0.2775*** (0.0018)	0.3831*** (0.0032)
Originator FE	Y	Y	Y	Y
State FE	Y	Y	Y	Y
Half-year FE	Y	Y	Y	Y
Observations	2,080,350	303,675	1,020,629	1,363,053
Adj. R ²	0.311	0.418	0.315	0.383

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. We look at different subsamples by borrowers incentive structures: primary or non-primary (columns (1) and (2)) and purchase or refinance (columns (3) and (4)). Variables are defined in Table 1. All continuous variables are winsorized at the 0.5 percent level and then standardized. All specifications include state, half-year, and originator fixed effects. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 5
Discontinuities for Low-documentation Loans Around the Credit Threshold in All Areas

	(1) All loan	(2) Reported second	(3) Misreported second
Panel A. Jump Ratio in Number of Loans			
Est. $620^+/620^-$	2.266	4.939	3.838
Panel B. Default Rate			
FICO ≥ 620 (β)	0.037	0.044	0.045
t-stat	(3.93)	(2.31)	(-1.15)

The table reports the discontinuities for low-documentation loans around a FICO score of 620 in all areas. Panel A presents the estimates from regressions where the dependent variable is the number of loans at each FICO score. Using local linear regressions of the RDD approach, we estimate the number of loans at FICO 620^- and FICO 620^+ and compute the ratio of the estimated number of loans at FICO 620^+ to 620^- . Panel B presents the estimates from regressions where the dependent variable takes a value of one if the loan becomes 90 days or more delinquent using the MBA method in the first three years after origination and zero otherwise. Using local linear regressions of the RDD approach with control variables, we estimate the difference in default rates between FICO 620^- and FICO 620^+ . We perform the estimation for all loans, loans with correctly reported simultaneous seconds, and loans with misreported simultaneous seconds, and the results are reported in columns (1) to (3), respectively. Bias-corrected t-values (standard errors clustered at the county level) following Calonico et al. (2014) are reported in parentheses.

Table 6

Discontinuities for Low-documentation Loans Around the Credit Threshold in High and Low Gambling Preference Areas

	(1) High	(2) Low	(3) High-Low
Panel A. Jump Ratio in Number of Loans (Est. 620+/620-)			
All loan	2.240	2.289	-0.049 (-0.52)
Reported second	5.498	4.581	0.917 (1.23)
Misreported second	2.763	4.826	-2.063 (-2.83)
Panel B. Default Rate			
All loan	0.036 (3.29)	0.037 (2.60)	-0.001 (-0.09)
Reported second	0.009 (0.21)	0.066 (2.87)	-0.056 (-1.46)
Misreported second	0.013 (0.16)	0.078 (-0.92)	-0.065 (-1.41)

The table reports the discontinuities for low-documentation loans around a FICO score of 620 separately in high and low gambling preference areas. Panel A presents the estimates from regressions where the dependent variable is the number of loans at each FICO score. Using local linear regressions of the RDD approach, we estimate the number of loans at FICO 620⁻ and FICO 620⁺ and compute the ratio of the estimated number of loans at FICO 620⁺ to 620⁻. Panel B presents the estimates from regressions where the dependent variable is the default dummy variable. Using local linear regressions of the RDD approach with control variables, we estimate the difference in default rates between FICO 620⁻ and FICO 620⁺. Using the diff-in-disc approach, we estimate the difference between high and low gambling preference areas. We perform the estimation for all loans, loans with correctly reported simultaneous seconds, and loans with misreported simultaneous seconds. The results for loans in high, low, and the difference between high and low gambling preference areas are reported in columns (1) to (3), respectively. For discontinuities, bias-corrected t-values (standard errors clustered at the county level) following Calonico et al. (2014) are reported in parentheses. For the difference between high and low gambling preference areas, t-statistics based on heteroskedasticity-consistent standard errors are reported in parentheses.

Table 7
Outcomes of Gambling Preference-Related Misrepresentation

	(1)	(3)	(3)
Misreported second	0.120*** (0.004)	0.120*** (0.004)	0.121*** (0.003)
CPRATIO		-0.001 (0.004)	-0.002 (0.004)
Misreported second*CPRATIO			0.014*** (0.002)
REL	-0.009*** (0.004)	-0.009*** (0.003)	-0.009*** (0.003)
Unemployment	0.006* (0.003)	0.006* (0.003)	0.006* (0.003)
HPA	0.046*** (0.004)	0.046*** (0.004)	0.046*** (0.004)
Education	0.004 (0.003)	0.004 (0.003)	0.004 (0.003)
Married	0.013*** (0.003)	0.013*** (0.003)	0.013*** (0.003)
Income	-0.012*** (0.002)	-0.012*** (0.002)	-0.012*** (0.002)
Urban	0.007* (0.004)	0.007* (0.004)	0.007* (0.004)
Total population	0.000 (0.006)	0.000 (0.006)	0.000 (0.006)
Over65	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)
Male-female ratio	-0.002 (0.003)	-0.002 (0.003)	-0.002 (0.003)
Minority	0.005 (0.004)	0.005 (0.004)	0.005 (0.004)
Interest rate	0.033*** (0.001)	0.033*** (0.001)	0.033*** (0.001)
FICO	-0.094*** (0.002)	-0.094*** (0.002)	-0.094*** (0.002)
Balance	0.023*** (0.002)	0.023*** (0.002)	0.023*** (0.002)
LTV	0.052*** (0.002)	0.052*** (0.002)	0.052*** (0.002)
ARM	0.040*** (0.002)	0.040*** (0.002)	0.040*** (0.002)
Negative amortization	0.037*** (0.004)	0.037*** (0.004)	0.037*** (0.004)
Option ARM	0.047*** (0.007)	0.047*** (0.007)	0.048*** (0.007)
Prepayment penalty	0.055*** (0.002)	0.055*** (0.002)	0.055*** (0.002)
Low or no doc.	0.054*** (0.002)	0.054*** (0.002)	0.054*** (0.002)
Cash-out	-0.025*** (0.002)	-0.025*** (0.002)	-0.025*** (0.002)
No-cash-out	0.010*** (0.004)	0.010*** (0.004)	0.010*** (0.004)
Investment	0.045*** (0.004)	0.045*** (0.004)	0.045*** (0.004)
Second-home	0.008** (0.003)	0.008** (0.003)	0.008** (0.003)
Reported second	0.142*** (0.007)	0.142*** (0.007)	0.141*** (0.007)
Originator FE	Y	Y	Y
State FE	Y	Y	Y
Half-year FE	Y	Y	Y
Observations	2,383,444	2,383,444	2,383,444
Adj. R ²	0.225	0.225	0.225

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan becomes 90 days or more delinquent using the MBA method in the first three years after origination and zero otherwise. All continuous variables are winsorized at the 0.5 percent level and then standardized. All specifications include state, half-year, and originator fixed effects. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 8
Alternative Gambling Preference Measures

	(1)	(2)	(3)	(4)
Gam. Ind. Emp.	0.002*** (0.001)		0.003*** (0.001)	
Gam. Ind. Est.		0.005** (0.002)		0.006** (0.002)
Casino Hotel Emp.			-0.001 (0.001)	
Casino Hotel Est.				0.002 (0.002)
Nonemployer Gam. Est.			0.001 (0.002)	0.001 (0.001)
REL	0.002 (0.001)	0.001 (0.001)	0.002 (0.001)	0.001 (0.001)
Unemployment	-0.001 (0.001)	-0.001 (0.001)	-0.000 (0.001)	-0.001 (0.001)
HPA	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.001 (0.002)
Education	0.001 (0.001)	-0.000 (0.001)	0.001 (0.001)	0.001 (0.001)
Married	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.002* (0.001)
Income	-0.001 (0.002)	-0.001 (0.002)	-0.001 (0.002)	-0.001 (0.002)
Urban	0.001 (0.002)	0.001 (0.002)	0.002 (0.001)	0.000 (0.001)
Total population	0.000 (0.001)	0.003* (0.002)	-0.000 (0.001)	0.004** (0.002)
Over65	-0.001** (0.001)	-0.002** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)
Male-female ratio	0.001 (0.001)	0.001 (0.001)	0.000 (0.001)	0.001 (0.001)
Minority	-0.002 (0.001)	-0.001 (0.001)	-0.002 (0.001)	-0.002 (0.001)
Interest rate	-0.008*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)
FICO	-0.006*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)
Balance	0.005*** (0.000)	0.005*** (0.000)	0.005*** (0.000)	0.005*** (0.000)
LTV	0.006*** (0.000)	0.006*** (0.000)	0.006*** (0.000)	0.006*** (0.000)
ARM	0.008*** (0.001)	0.008*** (0.001)	0.008*** (0.001)	0.008*** (0.001)
Negative amortization	-0.002 (0.002)	-0.002 (0.002)	-0.002 (0.002)	-0.002 (0.002)
Option ARM	-0.022*** (0.005)	-0.022*** (0.005)	-0.022*** (0.005)	-0.022*** (0.005)
Prepayment penalty	-0.009*** (0.002)	-0.009*** (0.002)	-0.009*** (0.002)	-0.009*** (0.002)
Low or no doc.	-0.008*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)
Cash-out	0.019*** (0.001)	0.019*** (0.001)	0.019*** (0.001)	0.019*** (0.001)
No-cash-out	0.016*** (0.001)	0.016*** (0.001)	0.016*** (0.001)	0.016*** (0.001)
Investment	0.027*** (0.001)	0.027*** (0.001)	0.027*** (0.001)	0.027*** (0.001)
Second-home	0.017*** (0.001)	0.017*** (0.001)	0.017*** (0.001)	0.017*** (0.001)
Simul. second	0.317*** (0.004)	0.317*** (0.004)	0.317*** (0.004)	0.317*** (0.004)
Originator FE	Y	Y	Y	Y
State FE	Y	Y	Y	Y
Half-year FE	Y	Y	Y	Y
Observations	1,050,588	1,050,588	1,050,588	1,050,588
Adj. R ²	0.318	0.318	0.318	0.318

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. Variables are defined in Table 1 while religiosity and the geographic controls are aggregated to CBSA level. All continuous variables are winsorized at the 0.5 percent level and then standardized. All specifications include state, half-year, and originator fixed effects. Standard errors clustered at the CBSA level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 9
Gambling Preference and Second-Lien Misrepresentation – Credit Expansion and Belief Distortion

	(1) High Income	(2) Low Income	(3) High FICO	(4) Low FICO	(5) High HPA	(6) Low HPA
CPRATIO	0.0008 (0.0005)	0.0013*** (0.0005)	0.0007 (0.0006)	0.0014*** (0.0005)	0.0012*** (0.0004)	0.0007 (0.0007)
REL	0.0002 (0.0004)	-0.0011** (0.0004)	-0.0007* (0.0004)	0.0003 (0.0004)	-0.0010*** (0.0004)	0.0004 (0.0004)
Unemployment	-0.0004 (0.0006)	-0.0009* (0.0006)	-0.0011* (0.0006)	-0.0001 (0.0005)	-0.0004 (0.0006)	-0.0002 (0.0007)
HPA	0.0025*** (0.0006)	-0.0005 (0.0007)	0.0021*** (0.0007)	0.0021*** (0.0008)	0.0006 (0.0007)	0.0054*** (0.0011)
Education	0.0003 (0.0003)	-0.0004 (0.0007)	-0.0002 (0.0004)	0.0007* (0.0004)	0.0003 (0.0004)	0.0005 (0.0004)
Married	-0.0002 (0.0004)	0.0004 (0.0006)	0.0003 (0.0005)	-0.0008** (0.0004)	0.0006 (0.0004)	-0.0004 (0.0004)
Income	0.0001 (0.0003)	-0.0017** (0.0006)	-0.0001 (0.0003)	-0.0008** (0.0004)	-0.0007** (0.0003)	-0.0000 (0.0003)
Urban	-0.0001 (0.0004)	-0.0006 (0.0004)	0.0003 (0.0005)	-0.0009** (0.0004)	-0.0008** (0.0004)	0.0002 (0.0004)
Total population	0.0006 (0.0005)	0.0018*** (0.0006)	0.0014** (0.0006)	0.0005 (0.0007)	0.0016*** (0.0004)	0.0008 (0.0006)
Over65	0.0006* (0.0003)	0.0000 (0.0004)	0.0002 (0.0003)	0.0004 (0.0003)	0.0002 (0.0004)	0.0007* (0.0003)
Male-female ratio	0.0003 (0.0003)	-0.0006* (0.0003)	0.0005 (0.0003)	-0.0004 (0.0003)	-0.0004 (0.0004)	0.0004 (0.0003)
Minority	0.0008 (0.0005)	-0.0000 (0.0006)	0.0007 (0.0006)	0.0002 (0.0005)	0.0005 (0.0006)	0.0006 (0.0006)
Interest rate	-0.0070*** (0.0004)	-0.0087*** (0.0006)	-0.0027*** (0.0004)	-0.0128*** (0.0005)	-0.0077*** (0.0004)	-0.0069*** (0.0006)
FICO	-0.0055*** (0.0004)	-0.0086*** (0.0005)	-0.0028*** (0.0004)	-0.0141*** (0.0006)	-0.0057*** (0.0004)	-0.0071*** (0.0007)
Balance	0.0044*** (0.0003)	0.0059*** (0.0003)	0.0047*** (0.0003)	0.0032*** (0.0004)	0.0050*** (0.0003)	0.0043*** (0.0003)
LTV	0.0052*** (0.0002)	0.0076*** (0.0005)	0.0026*** (0.0002)	0.0132*** (0.0006)	0.0055*** (0.0003)	0.0059*** (0.0004)
ARM	0.0092*** (0.0007)	0.0067*** (0.0007)	0.0093*** (0.0009)	0.0025*** (0.0005)	0.0068*** (0.0007)	0.0113*** (0.0008)
Negative amortization	-0.0042*** (0.0013)	-0.0040** (0.0017)	0.0070*** (0.0013)	-0.0087*** (0.0013)	-0.0036** (0.0014)	-0.0057*** (0.0017)
Option ARM	-0.0255*** (0.0049)	-0.0143** (0.0068)	-0.0305*** (0.0052)	-0.0084 (0.0051)	-0.0279*** (0.0060)	-0.0136** (0.0067)
Prepayment penalty	-0.0082*** (0.0009)	-0.0085*** (0.0009)	-0.0126*** (0.0010)	-0.0110*** (0.0010)	-0.0088*** (0.0010)	-0.0070*** (0.0011)
Low or no doc.	-0.0079*** (0.0005)	-0.0087*** (0.0008)	-0.0065*** (0.0006)	-0.0018** (0.0007)	-0.0079*** (0.0006)	-0.0080*** (0.0006)
Cash-out	0.0185*** (0.0006)	0.0179*** (0.0007)	0.0238*** (0.0007)	0.0176*** (0.0007)	0.0188*** (0.0007)	0.0177*** (0.0007)
No-cash-out	0.0158*** (0.0007)	0.0163*** (0.0011)	0.0166*** (0.0007)	0.0148*** (0.0011)	0.0157*** (0.0009)	0.0158*** (0.0008)
Investment	0.0256*** (0.0010)	0.0266*** (0.0013)	0.0235*** (0.0010)	0.0316*** (0.0008)	0.0252*** (0.0011)	0.0266*** (0.0013)
Second-home	0.0158*** (0.0010)	0.0137*** (0.0020)	0.0149*** (0.0008)	0.0196*** (0.0020)	0.0155*** (0.0010)	0.0160*** (0.0013)
Simul. second	0.3116*** (0.0019)	0.3192*** (0.0026)	0.2823*** (0.0025)	0.3581*** (0.0019)	0.3169*** (0.0025)	0.3088*** (0.0024)
Originator FE	Y	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y	Y
Half-year FE	Y	Y	Y	Y	Y	Y
Observations	1,833,682	550,259	1,324,342	1,059,862	1,371,294	1,012,419
Adj. R ²	0.315	0.329	0.298	0.357	0.321	0.315

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. We look at different subsamples related to credit expansion and belief distortion literature: high or low income (columns (1) and (2)), high or low credit score (columns (3) and (4)), and high or low HPA areas (columns (5) and (6)). The high or low income subsamples are defined by comparing local household median income with the sample median at the ZIP code level. The high or low credit score subsamples are divided by a FICO score of 670. The high and low HPA subsamples are defined based on the state-year median HPA, calculated from all available ZIP code-level data. All continuous variables are winsorized at the 0.5 percent level and then standardized. All specifications include state and year fixed effects. White robust standard errors are reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 10
Outcomes of Gambling Preference-Related Misrepresentation - Subsamples

	(1) High Income	(2) Low Income	(3) High FICO	(4) Low FICO	(5) High HPA	(6) Low HPA
Misreported second	0.118*** (0.003)	0.131*** (0.004)	0.094*** (0.003)	0.155*** (0.004)	0.128*** (0.004)	0.110*** (0.004)
CPRATIO	0.001 (0.004)	-0.010* (0.005)	-0.001 (0.004)	-0.005 (0.004)	0.001 (0.005)	0.001 (0.005)
Misreported second*CPRATIO	0.013*** (0.003)	0.019*** (0.003)	0.015*** (0.003)	0.015*** (0.003)	0.016*** (0.003)	0.012*** (0.004)
REL	-0.010*** (0.003)	-0.006 (0.004)	-0.009*** (0.003)	-0.007** (0.003)	-0.007* (0.004)	-0.013*** (0.004)
Unemployment	0.009*** (0.003)	0.005 (0.003)	0.009*** (0.003)	0.002 (0.003)	0.003 (0.004)	0.004* (0.002)
HPA	0.047*** (0.004)	0.048*** (0.006)	0.044*** (0.003)	0.054*** (0.005)	0.053*** (0.005)	0.052*** (0.008)
Education	0.005** (0.003)	0.001 (0.004)	0.003 (0.002)	0.005* (0.003)	0.003 (0.004)	0.003 (0.003)
Married	0.014*** (0.002)	0.013*** (0.004)	0.015*** (0.002)	0.008*** (0.003)	0.012*** (0.004)	0.014*** (0.003)
Income	-0.013*** (0.002)	-0.004 (0.003)	-0.012*** (0.001)	-0.010*** (0.003)	-0.011*** (0.003)	-0.016*** (0.002)
Urban	0.006 (0.004)	0.008* (0.004)	0.009** (0.004)	0.005 (0.004)	0.007 (0.004)	0.006 (0.005)
Total population	0.000 (0.006)	0.002 (0.007)	-0.001 (0.005)	-0.000 (0.007)	-0.004 (0.007)	0.005 (0.008)
Over65	0.005 (0.003)	-0.001 (0.004)	0.000 (0.003)	0.006** (0.003)	0.001 (0.003)	0.005* (0.003)
Male-female ratio	-0.002 (0.003)	-0.003 (0.003)	-0.002 (0.003)	-0.001 (0.003)	-0.002 (0.004)	-0.003 (0.003)
Minority	0.005 (0.004)	0.005 (0.005)	0.005 (0.003)	0.005 (0.004)	0.001 (0.005)	0.012*** (0.004)
Interest rate	0.034*** (0.001)	0.030*** (0.002)	0.040*** (0.002)	0.026*** (0.002)	0.030*** (0.001)	0.037*** (0.002)
FICO	-0.091*** (0.001)	-0.100*** (0.003)	-0.083*** (0.002)	-0.106*** (0.003)	-0.093*** (0.002)	-0.096*** (0.002)
Balance	0.019*** (0.001)	0.036*** (0.002)	0.016*** (0.002)	0.036*** (0.002)	0.025*** (0.002)	0.020*** (0.002)
LTV	0.050*** (0.002)	0.060*** (0.002)	0.041*** (0.002)	0.068*** (0.002)	0.059*** (0.002)	0.041*** (0.002)
ARM	0.041*** (0.002)	0.034*** (0.003)	0.059*** (0.002)	0.004 (0.003)	0.040*** (0.003)	0.039*** (0.002)
Negative amortization	0.037*** (0.003)	0.039*** (0.006)	0.044*** (0.005)	0.041*** (0.005)	0.029*** (0.005)	0.046*** (0.004)
Option ARM	0.054*** (0.009)	0.035*** (0.013)	0.104*** (0.012)	0.024*** (0.009)	0.045*** (0.009)	0.051*** (0.013)
Prepayment penalty	0.060*** (0.002)	0.041*** (0.003)	0.063*** (0.002)	0.036*** (0.003)	0.055*** (0.003)	0.056*** (0.003)
Low or no doc.	0.056*** (0.002)	0.050*** (0.002)	0.067*** (0.002)	0.056*** (0.002)	0.058*** (0.002)	0.049*** (0.002)
Cash-out	-0.017*** (0.002)	-0.049*** (0.003)	-0.010*** (0.003)	-0.049*** (0.002)	-0.023*** (0.003)	-0.026*** (0.003)
No-cash-out	0.019*** (0.003)	-0.025*** (0.005)	0.031*** (0.002)	-0.039*** (0.004)	0.013*** (0.003)	0.005 (0.005)
Investment	0.040*** (0.004)	0.059*** (0.005)	0.025*** (0.004)	0.084*** (0.005)	0.039*** (0.005)	0.052*** (0.005)
Second-home	0.010*** (0.003)	-0.003 (0.007)	0.012*** (0.003)	-0.001 (0.005)	0.006 (0.005)	0.009*** (0.003)
Reported second	0.138*** (0.006)	0.155*** (0.009)	0.116*** (0.007)	0.192*** (0.007)	0.160*** (0.008)	0.117*** (0.005)
Originator FE	Y	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y	Y
Half-year FE	Y	Y	Y	Y	Y	Y
Observations	1,832,860	549,841	1,324,100	1,058,863	1,370,589	1,011,884
Adj. R ²	0.225	0.215	0.224	0.181	0.237	0.208

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. We look at different subsamples related to credit expansion and belief distortion literature: high or low income (columns (1) and (2)), high or low credit score (columns (3) and (4)), and high or low HPA areas (columns (5) and (6)). The high or low income subsamples are defined by comparing local household median income with the sample median at the ZIP code level. The high or low credit score subsamples are divided by a FICO score of 670. The high and low HPA subsamples are defined based on the state-year median HPA, calculated from all available ZIP code-level data. All continuous variables are winsorized at the 0.5 percent level and then standardized. All specifications include state, half-year, and originator fixed effects. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 11
Alternative Default Measures for Loan Performance Tests

	(1) 90+ delinq in 3 year	(2) 90+ delinq in 2 year	(3) 60+ delinq in 3 year	(4) B/F/REO in 3 year
Misreported second	0.121*** (0.003)	0.066*** (0.003)	0.123*** (0.003)	0.109*** (0.003)
CPRATIO	-0.002 (0.004)	-0.002 (0.002)	-0.001 (0.004)	-0.003 (0.003)
Misreported second*CPRATIO	0.014*** (0.002)	0.007*** (0.002)	0.014*** (0.003)	0.013*** (0.003)
REL	-0.009*** (0.003)	-0.005*** (0.002)	-0.009*** (0.003)	-0.007*** (0.003)
Unemployment	0.006* (0.003)	0.006*** (0.002)	0.004 (0.003)	0.007** (0.003)
HPA	0.046*** (0.004)	0.025*** (0.003)	0.046*** (0.004)	0.037*** (0.003)
Education	0.004 (0.003)	0.002 (0.001)	0.003 (0.003)	0.003 (0.002)
Married	0.013*** (0.003)	0.007*** (0.001)	0.013*** (0.003)	0.009*** (0.002)
Income	-0.012*** (0.002)	-0.008*** (0.001)	-0.012*** (0.002)	-0.009*** (0.002)
Urban	0.007* (0.004)	0.004* (0.002)	0.006 (0.004)	0.006* (0.003)
Total population	0.000 (0.006)	0.001 (0.003)	0.000 (0.006)	-0.001 (0.005)
Over65	0.003 (0.003)	0.003 (0.002)	0.003 (0.003)	0.005 (0.003)
Male-female ratio	-0.002 (0.003)	-0.001 (0.002)	-0.002 (0.003)	-0.002 (0.002)
Minority	0.005 (0.004)	0.003 (0.002)	0.006 (0.004)	0.001 (0.003)
Interest rate	0.033*** (0.001)	0.047*** (0.001)	0.027*** (0.001)	0.031*** (0.001)
FICO	-0.094*** (0.002)	-0.068*** (0.001)	-0.113*** (0.002)	-0.058*** (0.001)
Balance	0.023*** (0.002)	0.020*** (0.001)	0.020*** (0.002)	0.019*** (0.002)
LTV	0.052*** (0.002)	0.027*** (0.001)	0.055*** (0.002)	0.043*** (0.001)
ARM	0.040*** (0.002)	0.039*** (0.002)	0.033*** (0.002)	0.045*** (0.002)
Negative amortization	0.037*** (0.004)	0.046*** (0.003)	0.034*** (0.004)	0.023*** (0.005)
Option ARM	0.048*** (0.007)	0.028*** (0.006)	0.051*** (0.008)	0.036*** (0.006)
Prepayment penalty	0.055*** (0.002)	0.019*** (0.001)	0.061*** (0.002)	0.025*** (0.002)
Low or no doc.	0.054*** (0.002)	0.031*** (0.001)	0.058*** (0.002)	0.051*** (0.002)
Cash-out	-0.025*** (0.002)	-0.025*** (0.002)	-0.020*** (0.002)	-0.026*** (0.002)
No-cash-out	0.010*** (0.004)	0.001 (0.003)	0.014*** (0.003)	0.009*** (0.003)
Investment	0.045*** (0.004)	0.028*** (0.003)	0.044*** (0.004)	0.058*** (0.003)
Second-home	0.008** (0.003)	0.001 (0.002)	0.009** (0.004)	0.026*** (0.003)
Reported second	0.141*** (0.007)	0.079*** (0.005)	0.144*** (0.007)	0.113*** (0.006)
Originator FE	Y	Y	Y	Y
State FE	Y	Y	Y	Y
Half-year FE	Y	Y	Y	Y
Observations	2,383,444	2,383,444	2,383,444	2,383,444
Adj. R ²	0.225	0.156	0.236	0.155

The table shows OLS estimates from regressions in which the dependent variable uses different measures of default. Column (1) uses 90 days or more delinquency using MBA method in the first three years after origination (baseline). Column (2) uses 90 days or more delinquency using MBA method in the first two years after origination. Column (3) uses 60 days or more delinquency using MBA method in the first two years after origination. Column (4) uses bankruptcy/foreclosure/REO in the first three years after origination. All continuous variables are winsorized at 0.5 percent level and then standardized. All specifications include state, half-year, and originator fixed effects, and standard errors clustered at the county level are reported in the parentheses.

* $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 12
Gambling Preference and Mortgage Misrepresentation - Probit Model

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
CPRATIO	0.006** (0.002)	0.014*** (0.003)	0.015*** (0.002)	0.012*** (0.002)	0.006*** (0.001)	0.006** (0.003)	0.004** (0.002)
Geographic controls	N	Y	N	Y	Y	Y	Y
Loan chars controls	N	N	Y	Y	Y	Y	Y
Originator FE	N	N	N	N	Y	Y	Y
State FE	N	N	N	N	N	Y	Y
Half-year FE	N	N	N	N	N	N	Y
Observations	638,544	609,627	626,978	598,574	478,730	470,402	459,341
Pseudo R ²	0.000	0.005	0.024	0.026	0.223	0.221	0.258

The table shows the marginal effects of CPRATIO from probit models where the dependent variable takes a value of one if the loan is misreported and zero otherwise. All continuous variables, except for the crime rates, are winsorized at the 0.5 percent level and then standardized. All specifications include state, half-year, and originator fixed effects. Standard errors clustered at the county level are reported in parentheses. $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

Table 13
Gambling Preference and Mortgage Misrepresentation - Causal Forest

	(1)
	Misreported second
CPRATIO	0.0013** (0.0006)

The table shows causal forest estimates from regressions where the dependent variable takes a value of one if the loan is misreported and zero otherwise. All control variables in Table 3 are used for growing trees and forests. Before growing the trees, all continuous variables are winsorized at the 0.5 percent level and then standardized. A half-year variable is created, starting from the first half of 2005 as 1, and used as a variable in growing trees and forests. Observations are clustered by state, and each unit is given the same weight (so that larger clusters receive more weight). 2000 trees are grown in the causal forest. The fraction used for determining splits (honesty) is 50 percent. Standard errors are reported in parentheses. $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

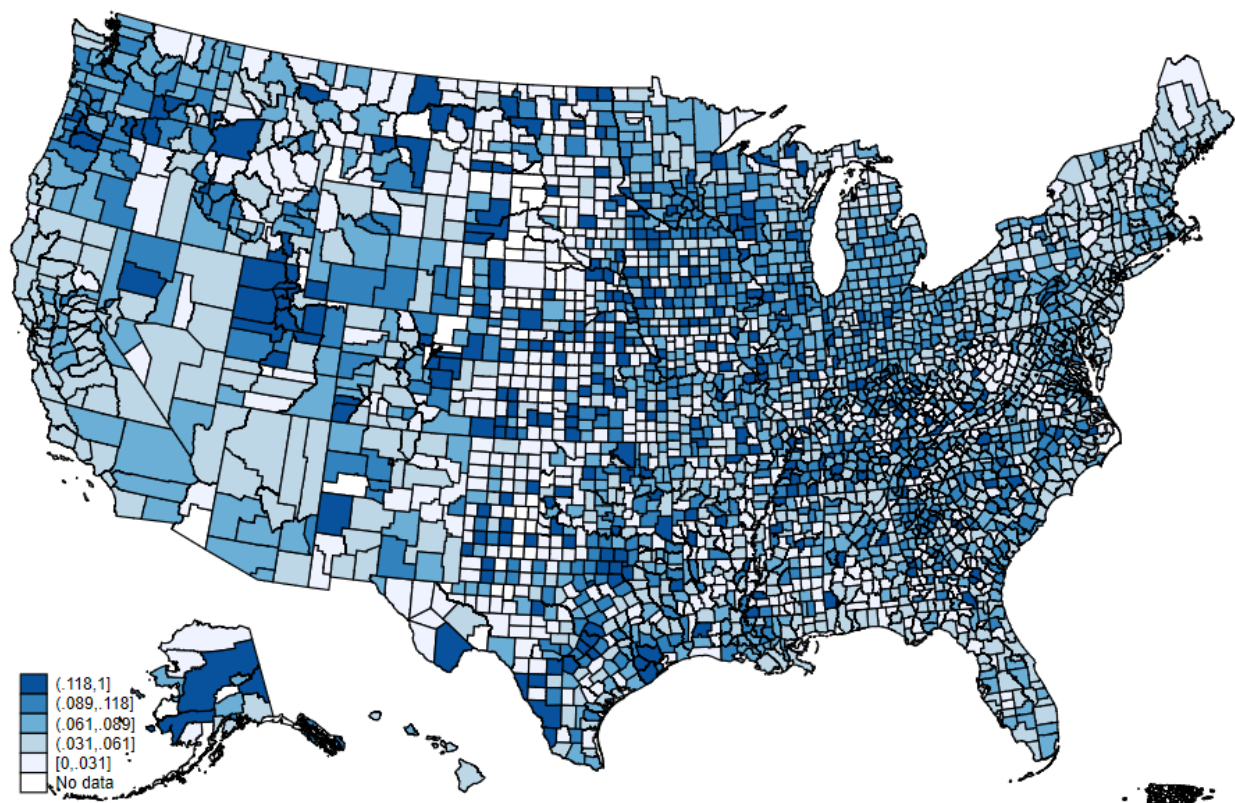


Figure 1
Second-lien Misrepresentations Distribution
The figure plots the county level proportion of second-lien misrepresentation in all loans.

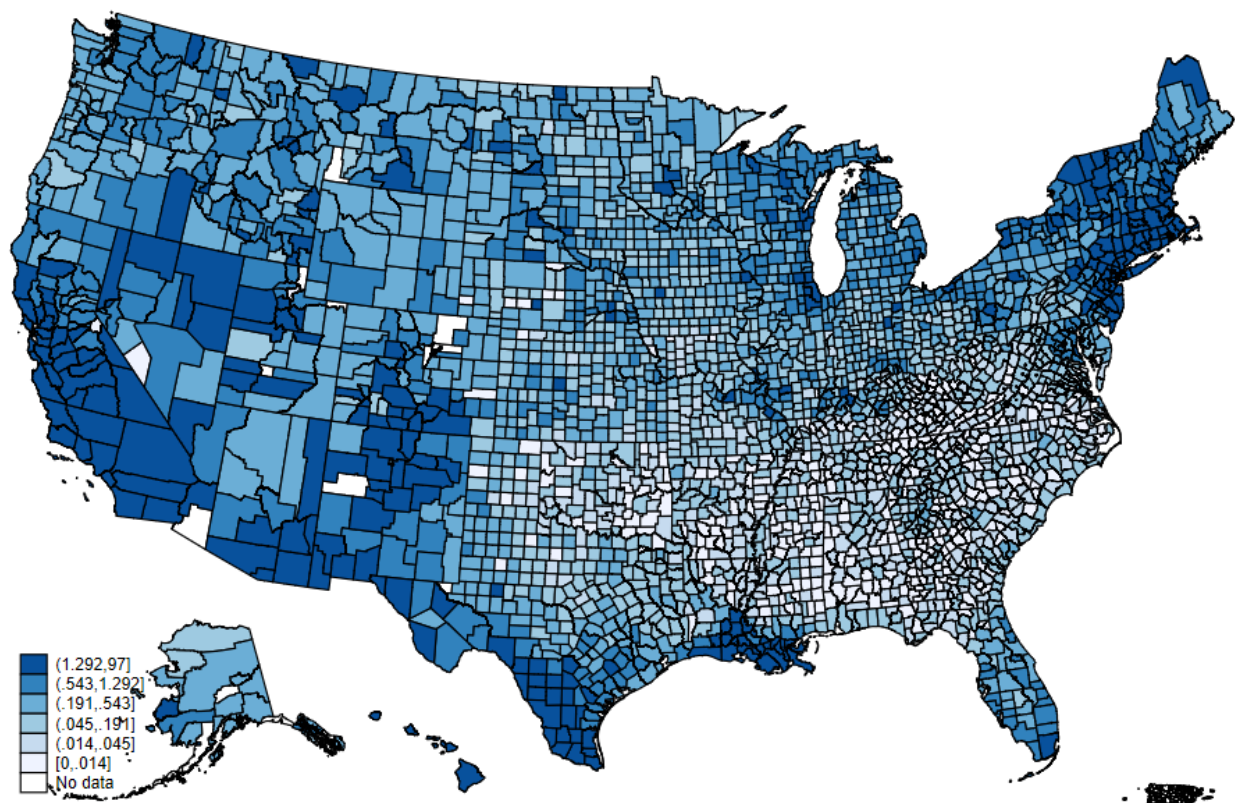


Figure 2
CPRATIO Distribution
The figure plots the county level CPRATIO across the US.

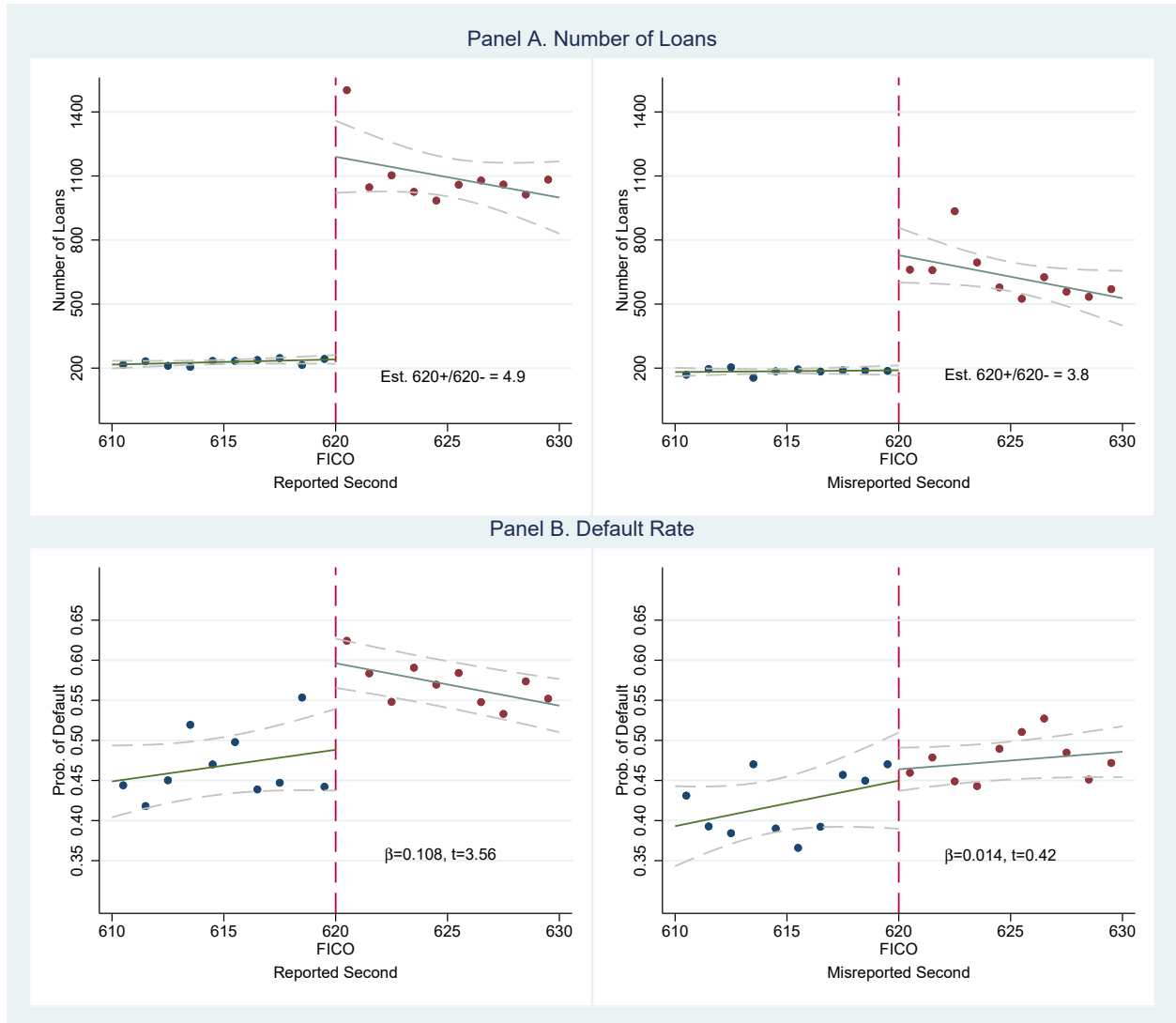


Figure 3
Discontinuities for Low-Documentation Loans with Simultaneous Seconds Around Credit Score Thresholds

The figure plots discontinuities for low-documentation loans with simultaneous seconds by credit score. Panel A reports the number of loans, and Panel B reports the default rate. The circles represent the average number of loans or probability of default for each credit score between 610 and 629. The solid lines show the linear fit on either side of the credit score threshold, with dashed lines denoting the 95% confidence interval. The left subfigures report results for loans with correctly reported simultaneous seconds, and the right subfigures present results for loans with misreported simultaneous seconds. As shown, the jump ratio in the number of loans is smaller for misreported seconds than for correctly reported seconds. Additionally, the jump in default rates is large and significant for correctly reported seconds but small and insignificant for misreported seconds.

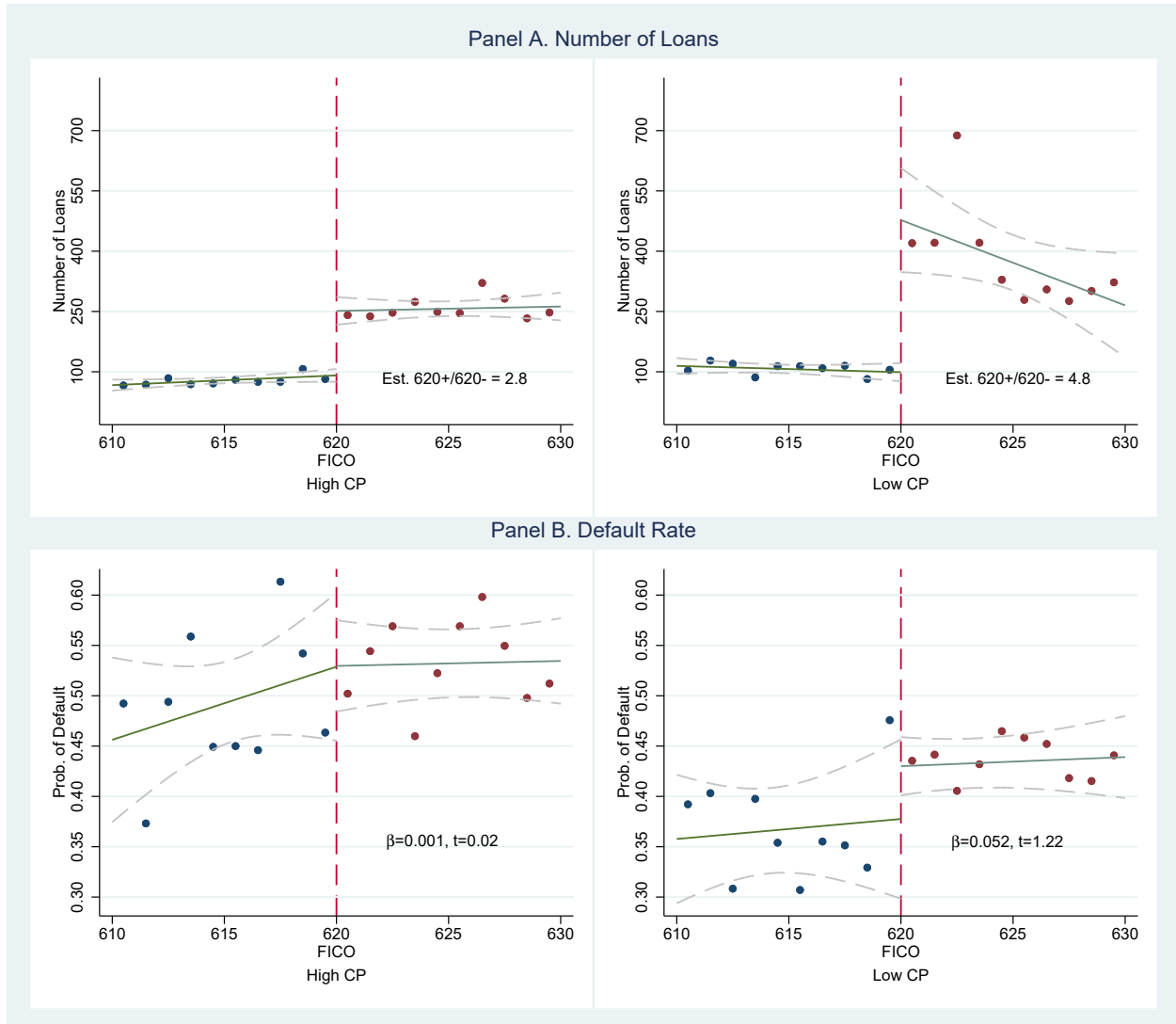


Figure 4
Discontinuities for Low-Documentation Loans with Misreported Simultaneous Seconds Around Credit Score Thresholds by Gambling Preference

The figure plots discontinuities for low-documentation loans with misreported simultaneous seconds by credit score. Panel A reports the number of loans, and Panel B reports the default rate. The circles represent the average number of loans for each credit score between 610 and 629. The solid lines show the linear fit on either side of the credit score threshold, with dashed lines denoting the 95% confidence interval. The left subfigures report results for loans in high gambling preference areas, while the right subfigures present results for loans in low gambling preference areas. As shown, the jump ratio is much smaller in high gambling preference areas than in low gambling preference areas. Additionally, the jump in default rates is insignificant in both groups, though larger in magnitude in low gambling preference areas.

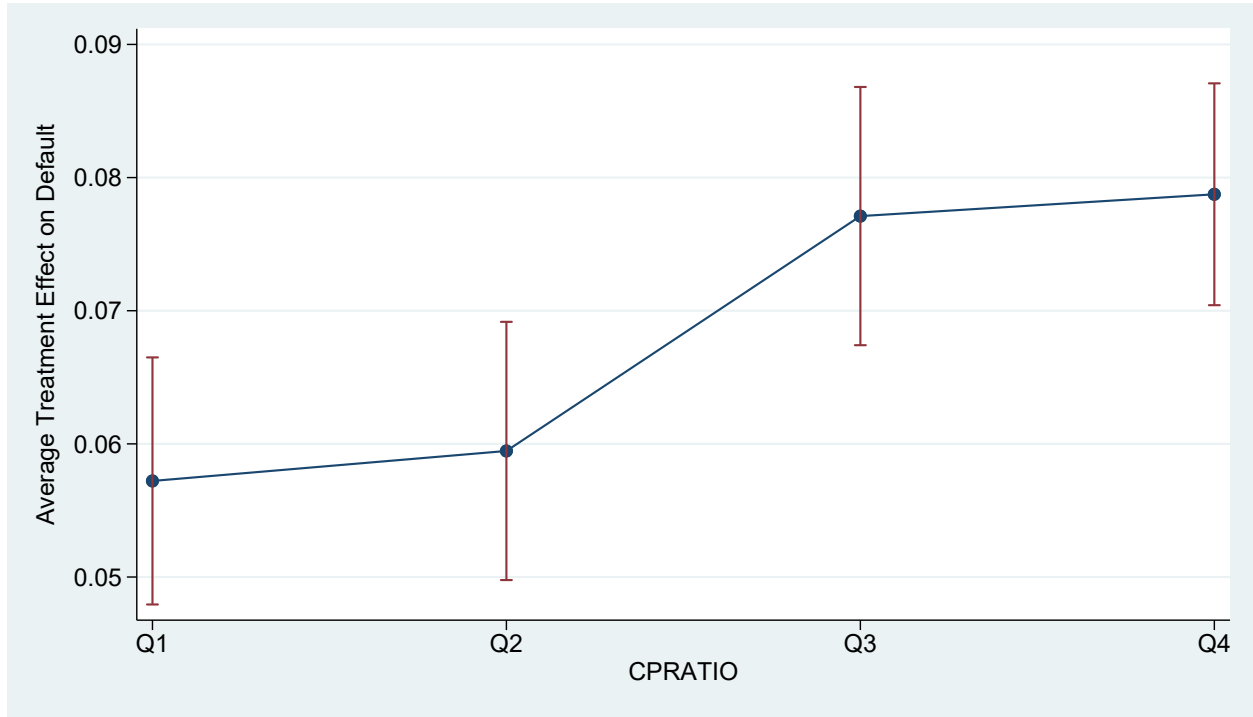


Figure 5
Heterogeneous Treatment Effects of Second-lien Misrepresentation on Default Conditioned on Different Levels of Gambling Preference

The figure plots the heterogeneous treatment effects of second-lien misrepresentation on default, conditioned on different levels of gambling preference, using the causal forest approach. The dependent variable is an indicator that takes a value of one if the loan becomes 90 days or more delinquent using the MBA method in the first three years after origination. The average treatment effect of mortgage misrepresentation is estimated at different levels of local gambling preference (four quarters divided by CPRATIO). All control variables in column (3) of Table 7 are used for growing trees and forests. Before growing the trees, all continuous variables are winsorized at the 0.5 percent level and then standardized. A half-year variable is created, starting from the first half of 2005 as 1, and used as a variable in growing trees and forests. Observations are clustered by state, and each unit is given the same weight (so that larger clusters receive more weight). 2000 trees are grown in the causal forest. The fraction used for determining splits (honesty) is 50 percent.

Appendix

A Alternative Gambling Preference Measures

In this section, we provide detailed information regarding the use of alternative gambling preference measures.

A.1 Data and Measure Construction

Since CPRATIO may capture broader religious influences, such as attitudes toward forgiveness and dishonesty, we also employ more direct measures of local gambling preference. Specifically, we use per capita employment in the gambling industry and per capita number of gambling establishments to quantify local gambling engagement.

To construct these measures, we extract employment and number of establishments data at the core-based statistical area (CBSA) level¹ for the years 2005 through 2007 from the County Business Patterns (CBP) dataset provided by the U.S. Census Bureau. To identify gambling-specific industries, we retain observations with NAICS code 7132, which includes 71321 for Casinos (except Casino Hotels) and 71329 for Other Gambling Industries. These counts are then normalized by dividing by the total population of the area. Population figures for 2005 to 2007 are linearly interpolated based on CBSA-level data from the 2000 and 2010 Census.

In addition to the two gambling preference measures described above, we construct supplementary proxies. Because CBP classifies businesses based on their primary services, casino hotels (NAICS code 72110) are excluded from the gambling industry category. However, as these businesses also provide gambling services to the public, we extract employment and establishment data for casino hotels separately, normalizing them by total population as well.

Since the CBP dataset includes only employer-based businesses, some small gambling venues without formal employees may be omitted. To account for these, we incorporate data from the Nonemployer Statistics (NES) datasets, also maintained by the U.S. Census Bureau. We retain NES observations with NAICS code 7132². The number of nonemployer gambling establishments is likewise normalized by CBSA population.

¹This refers to the MSA File in the CBP dataset; however, the MSA identification codes correspond to CBSA codes.

²During our sample period, no casino hotel reports having employees; therefore, NAICS code 72110 does not appear in the NES data.

While the CBP dataset records the number of establishments without restrictions, the employment data suffers from a key limitation: many cells are suppressed to preserve confidentiality (Eckert et al., 2021)³. To ensure sufficient CBSA-level coverage, we apply the following method to approximate employment figures⁴. First, for each CBSA in which a gambling-related NAICS code appears, if the employment value is nonzero, we retain it as is. Second, for cases where the employment value is reported as zero, we estimate it by multiplying the number of establishments in each employment-size interval by the median value of that interval. For instance, in CBSA 11460, there are two establishments, one in the 1–4 interval and one in the 5–9 interval, so the estimated employment is calculated as $1 \times 2.5 + 1 \times 7 = 9.5$. For establishments falling within the highest interval (i.e., open-ended, 5,000 or more), we conservatively assign the lower bound value of 5,000. Third, we compare the estimated employment figure with the lower bound of the corresponding employment level flag and retain the value that is greater⁵.

The NES dataset also contains limitations: numerous cells are suppressed to preserve confidentiality or to avoid quality issues. In such cases, the number of establishments is recorded as zero; however, the establishment flags indicate that the zero does not reflect true nonexistence. Instead, these suppressed values result from either unwillingness to disclose individual business information or concerns regarding data reliability. To address this issue, we implement the following imputation strategy. For observations with flag D (confidentiality concerns), we replace the zero with a value of 2⁶. For observations with flag S (reliability concerns), we substitute the zero with the state-level average for the corresponding industry category.

For geographic control variables, the U.S. Census Bureau does not provide CBSA-level data for the year 2000. To address this limitation, we aggregate county-level data up to the CBSA level. For ratio variables, we first aggregate the numerator and denominator separately across constituent counties, and then compute the ratio based on the aggregated values. For unemployment rates obtained from the U.S. Bureau of Labor Statistics, we apply

³Another limitation discussed in Eckert et al. (2021) is the periodic reclassification of industries. However, since changes to gambling-related NAICS codes are infrequent and our sample period is short, this concern is minimal for our analysis.

⁴Eckert et al. (2021) proposes a solution for county-level data, but not for CBSA-level data. We adapt their core approach in a simplified form due to constraints that are difficult to satisfy at the CBSA level. Our primary strategy involves using the median value within employment intervals and verifying that the estimated employment remains within the bounds of the corresponding employment level flag interval.

⁵Refer to Tables 1 and 3 in Eckert et al. (2021) for definitions of employment intervals and employment level flag intervals, respectively. In our sample, all estimated values fall below the corresponding upper bounds of the employment level flags.

⁶We use 2 since the smallest observable nonzero value is 3. In untabulated results, substituting with 1 does not materially affect our findings.

the same aggregation procedure to the annual average data and implement a one-year lag in our regressions. For HPA, we use the CBSA-level HPI data directly from the FHFA.

Table A1 presents the summary statistics for all constructed gambling preference measures.

A.2 Additional Tests

To validate our alternative measures and explore their relation with CPRATIO, we do the following tests.

First, to examine the relationship between CPRATIO and both the alternative gambling preference measures and three additional control variables, we conduct a univariate sorting test. CBSAs are sorted into quintiles according to their CPRATIO values, and we calculate equal-weighted quintile averages for the two alternative measures of gambling preference as well as the three control variables. Table A2 summarizes the results. For the two alternative measures, we find that gambling industry employment tends to increase with CPRATIO⁷, and the difference between the highest and lowest quintiles is statistically significant. This suggests that employment in the gambling industry captures a construct that is consistent with CPRATIO. In contrast, the relationship between gambling industry establishments and CPRATIO is hump-shaped, with the High quintile displaying slightly elevated values. This pattern may reflect legal constraints. For example, both a lottery venue and a casino venue are counted as a single establishment, even though casinos typically offer a broader array of gambling services. Moreover, lotteries are generally more easily approved due to their association with public funding for education and healthcare, whereas casinos face stricter legal barriers. As a result, in regions with low gambling preference where lotteries are permitted but casinos are prohibited, the number of recorded establishments may increase. Conversely, in areas where casinos are allowed, fewer establishments may exist, but each offers stronger access to gambling activities.

For the three control variables, employment and establishment counts in casino hotels do not show a substantial increase with CPRATIO. This outcome may stem from two factors. First, the presence of casino hotels is influenced not only by local gambling demand but also by tourist-oriented gambling activity. As a result, local gambling preference does not necessarily correlate with casino hotel employment or establishment density. Second, exceptions in gambling legality, such as tribal casinos, permit casino hotel operations in certain regions even when local gambling preference is low. In contrast, the number of nonemployer

⁷It is reasonable that gambling industry employment and CPRATIO do not exhibit a strictly monotonic increasing relation, since Catholics, while generally more tolerant of gambling activities, often oppose the excessive expansion of gambling establishments because of their potential social harms.

gambling establishments tends to increase with CPRATIO. This pattern suggests that such establishments complement minor local gambling demand and provide an additional proxy for community-level gambling activity.

Second, to further examine whether CPRATIO can explain variation in the alternative gambling preference measures and the three additional control variables, we estimate a series of OLS regressions. Each dependent variable is regressed on CPRATIO and CBSA-level geographic control variables with state and year fixed effects. All continuous variables are winsorized at the 0.5 percent level and standardized prior to analysis. Table A3 presents the regression results. For the two alternative measures, gambling industry employment and establishment, CPRATIO is positively and significantly associated with both outcomes, indicating that higher CPRATIO values correspond to greater levels of these supply-side indicators. This supports the use of CPRATIO as a proxy for local gambling preference. In contrast, results for casino hotels are mixed. While the per capita number of establishments is positively correlated with CPRATIO, the per capita employment in the sector is negatively correlated. This divergence reinforces our decision to treat casino hotel variables as controls for tourist-oriented gambling demand rather than proxies for local gambling preference. Finally, for nonemployer gambling establishments, which serve as a complementary measure for minor local gambling activity, we also observe a significantly positive coefficient on CPRATIO. This suggests that higher CPRATIO values are associated with increased presence of nonemployer gambling businesses, further corroborating CPRATIO’s relevance as an indicator of local gambling preference.

A.3 Additional Illustration

In this section, we address several concerns related to the use of alternative measures of local gambling preference.

First, several limitations prevent us from using the alternative measures as the basis for our primary analysis. Although employment and establishment data in the gambling industry may more directly reflect local gambling preferences, these indicators represent supply-side dynamics rather than demand-side behavior. Furthermore, they are influenced by factors such as local legality and economic conditions, which introduces additional noise into the measurement. Another key limitation is the restricted coverage. These alternative measures span only 460 CBSAs, encompassing approximately 1,187 counties, and cover around 1 million loans. In contrast, CPRATIO includes data for nearly 3,000 counties and covers almost the entire sample of 3 million loans. This broader coverage is especially critical when analyzing lender-specific effects and borrower preference heterogeneity.

Second, we rely on CBSA-level data rather than county-level data to construct our gambling preference measures. The rationale for using a broader geographic and economic unit is that residents in most U.S. metropolitan areas can easily commute within the city. Consequently, individuals residing in counties with few gambling establishments may work or gamble in adjacent counties with greater access. As a result, low employment or establishment counts in a given county do not necessarily reflect low gambling preference among its residents. By contrast, CBSAs are sufficiently large that cross-CBSA commuting for work or gambling is relatively rare. Instances of inter-CBSA gambling, such as through tourism, are mostly captured by the casino hotel measures. Therefore, the CBSA level offers a more robust and stable proxy for local gambling preferences than county-level data.

Third, we restrict our sample to CBSAs where establishments exist in all three categories: gambling industry, casino hotels, and nonemployer gambling businesses. This selection improves the clarity and reliability of our gambling preference measures. In such areas, false zeros, resulting from legal prohibitions, data suppression, or missing supply, are minimized. In untabulated regressions, when we replace missing values with zeros and include all CBSAs, the coefficient on the gambling preference measure remains positive but becomes only marginally significant. Furthermore, when we split the sample into high and low per capita establishment groups, the coefficient is strongly significant in the high-establishment group but insignificant in the low group. This pattern suggests that second-lien misrepresentation does rise with elevated local gambling preference, whereas low values in the measure may reflect extraneous noise rather than genuinely low gambling tendencies.

Table A1
Descriptive Statistics for Alternative Gambling Preference Measures

	N	Mean	SD	P10	P25	P50	P75	P90
Gambling Industries (NAICS 7132, CBSA-year level)								
Employment (per 10,000 residents)	1,201	19.60	42.74	0.22	0.69	2.53	16.41	59.50
Establishment (per 10,000 residents)	1,201	0.39	0.85	0.03	0.06	0.14	0.29	0.76
Casino Hotel (NAICS 72110, CBSA-year level)								
Employment (per 10,000 residents)	258	118.49	264.87	0.07	0.50	16.90	77.41	389.73
Establishment (per 10,000 residents)	258	0.24	0.60	0.00	0.01	0.07	0.21	0.55
Nonemployer Gambling (NAICS 7132, CBSA-year level)								
Establishment (per 10,000 residents)	1,239	0.38	0.50	0.00	0.14	0.27	0.45	0.76

The table presents descriptive statistics for the alternative local gambling preference measures and its related controls, covering the sample period from 2005 to 2007. We report the number of observations, mean, standard deviation, 25th percentile, 50th percentile, and 75th percentile.

Table A2
Univariate Sorting

	Gambling Industries		Casino Hotel		Nonemployer Gambling
Quintile	Employment	Establishment	Employment	Establishment	Establishment
Low	16.26	0.26	114.20	0.18	0.28
Q2	11.77	0.47	140.31	0.79	0.35
Q3	18.67	0.57	49.83	0.11	0.43
Q4	26.47	0.34	95.67	0.21	0.40
High	24.92	0.31	219.91	0.25	0.46
High-Low	8.66	0.05	105.72	0.08	0.18
t-Statistic	2.02	0.78	1.72	1.10	4.01

The table presents univariate sorting results for five gambling preference related measures: per capita employment in the gambling industry (Gambling Industries Employment), per capita number of gambling establishments (Gambling Industries Establishment), per capita employment in the casino hotel industry (Casino Hotel Employment), per capita number of casino hotel establishments (Casino Hotel Establishment), and per capita number of nonemployer gambling establishments (Nonemployer Gambling Establishment). CBSAs are sorted into quintiles based on CPRATIO, and for each quintile, the equal-weighted average of the five gambling preference related measures is computed. The table reports White robust t-statistics for the difference in mean values between the High and Low CPRATIO quintiles.

Table A3
Alternative measures estimate

	Gambling Industries		Casino Hotel		Nonemployer Gambling
	Employment	Establishment	Employment	Establishment	Establishment
CPRATIO	0.112*** (0.043)	0.031* (0.019)	-0.194*** (0.068)	0.194* (0.113)	0.134*** (0.028)
REL	-0.116** (0.059)	-0.053 (0.042)	-0.084 (0.147)	-0.374* (0.206)	-0.179*** (0.061)
Unemployment	-0.013 (0.061)	-0.080*** (0.022)	-0.137 (0.090)	-0.100 (0.072)	0.027 (0.072)
HPA	0.008 (0.045)	-0.039 (0.026)	0.108 (0.083)	-0.030 (0.076)	0.075 (0.046)
Education	0.018 (0.058)	-0.073** (0.033)	-0.703*** (0.214)	0.272 (0.198)	-0.084 (0.063)
Married	0.059 (0.047)	-0.001 (0.031)	-0.480*** (0.150)	0.252 (0.153)	-0.017 (0.048)
Income	-0.206*** (0.069)	-0.054* (0.031)	0.950*** (0.250)	-0.292 (0.253)	0.162** (0.065)
Urban	0.035 (0.063)	0.070** (0.032)	-0.397*** (0.102)	0.037 (0.129)	-0.008 (0.044)
Total population	-0.138*** (0.050)	-0.132*** (0.025)	-0.023 (0.119)	-0.294*** (0.102)	-0.063 (0.051)
Over65	0.113*** (0.043)	0.025 (0.027)	0.068 (0.110)	-0.107 (0.110)	0.038 (0.044)
Male-female ratio	0.074 (0.050)	0.011 (0.017)	0.027 (0.080)	-0.081 (0.074)	-0.067** (0.026)
Minority	0.152** (0.063)	0.008 (0.020)	0.169 (0.110)	0.173** (0.087)	-0.032 (0.042)
State FE	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Observations	1,118	1,118	240	240	1,147
Adj. R2	0.307	0.723	0.541	0.408	0.262

The table shows OLS estimates from regressions where the dependent variables are the alternative gambling preference measures and related control variables. Religiosity and the geographic controls are aggregated to CBSA level. All continuous variables are winsorized at the 0.5 percent level and then standardized. All specifications include state and year fixed effects. White robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

B Local Tolerance for Law-breaking Behaviors

Since second-lien misrepresentation is a form of misrepresentation that violates the law, one concern is that our findings could be driven by our gambling preference measure inadvertently capturing variation in the general local tolerance for law-breaking behavior. To address this possibility, we include additional controls for both Part I and Part II crimes as categorized by the Uniform Crime Reporting (UCR) Program.

In this section, we provide detailed information on the crime rate data and presents results with these crime rate control variables.

B.1 Data and Measures

We collect county-level crime data for the years 2005 to 2007 from the National Neighborhood Data Archive (NaNDA) and the Uniform Crime Reporting (UCR) Program Data available through the ICPSR website.

We rely on NaNDA primarily for data on Part I offenses. From NaNDA, we extract county population used to calculate county-level crime reports, county population used to calculate arrests, total crimes including arsons⁸, total murders, total rapes, total robberies, total aggravated assaults, total burglaries, total larcenies, total auto thefts, and total arsons. Following the dataset definitions, we classify violent crimes as the sum of murders, rapes, robberies, and aggravated assaults, and property crimes as the sum of burglaries, larcenies, and auto thefts. We compute crime rates by dividing the number of incidents by the county population used to calculate county-level crime reports and multiplying by 10,000, yielding crime rates expressed as the number of crimes per 10,000 residents per year.

We use UCR Program Data primarily for Part II offenses. From this source, we obtain counts for nearly all types of Part II offenses⁹, such as fraud, embezzlement, and drug abuse violations. We aggregate all such counts to obtain the total number of Part II crimes. Additionally, following the Denver UCR Crime Types guide, we define against person crimes (as the sum of other assault, sex offenses, and offenses against the family and children), public disorder crimes (as the sum of vandalism, prostitution and commercialized vice, drunkenness, and disorderly conduct), drug and alcohol offenses (as the sum of drug abuse violations and liquor law offenses), white collar crimes (as the sum of forgery and counterfeiting, fraud, and embezzlement), and all other crimes (as the sum of weapons, stolen property, and all other offenses)¹⁰. To calculate Part II crime rates, we divide the number of incidents by the

⁸This count covers all Part I offenses and is used to calculate the Part I crime rate.

⁹Curfew and runaways are excluded because they apply only to juveniles.

¹⁰Although gambling and driving under influence are not belong to any above categories in the guide, we

county population used to calculate arrests and multiply by 10,000, yielding rates per 10,000 residents per year.¹¹ Table B1 presents summary statistics for all crime rates.

B.2 Results for Granular Breakdown of the Crime Categories

We next examine whether controlling for Part I and Part II crime rates, along with their specific subcategories, affects the results. Columns (1) and (2) include controls for Part I crimes and their subcategories, respectively. Columns (3) and (4) include controls for Part II crimes and their subcategories, respectively. Columns (5) and (6) include controls for both Part I and Part II crimes and for all subcategories, respectively.

Table B2 reports the regression results. The coefficient of CPRATIO remains statistically significant and of similar economic magnitude after adding these crime controls. This indicates that our findings are not primarily driven by variation in local tolerance for law-breaking behavior.

include them into the total Part II offenses.

¹¹We use the county population for arrests as the denominator because arrest data are the only counts available for Part II offenses. Using this denominator ensures consistency. Note that the county population used for arrests may differ slightly from that used for county-level crime reports due to differences in distributing populations across multi-county jurisdictions. Further details are provided in the UCR Program Guide.

Table B1
Descriptive Statistics for Crime Rates

	N	Mean	SD	P10	P25	P50	P75	P90
Crime Rate (per 10,000 residents per year)								
Part I Crime	8,829	271.33	172.67	85.10	153.88	241.80	356.59	500.00
Violent	8,829	28.70	26.78	5.40	11.15	21.61	38.31	60.67
Property	8,829	240.83	152.90	74.07	136.02	215.04	317.43	442.95
Arson	8,829	1.73	2.80	0.00	0.00	1.20	2.45	4.01
Part II Crime	9,411	364.16	244.87	84.75	211.96	342.33	482.39	632.57
Against person	9,411	47.02	35.00	7.76	24.20	41.65	62.80	89.13
Public disorder	9,411	46.74	43.80	2.81	16.60	35.93	64.26	101.67
Drug & alcohol	9,411	71.46	56.75	12.78	36.04	62.95	93.54	132.63
White collar	9,411	16.60	23.77	0.00	3.44	9.04	20.70	39.64
All others	9,411	132.58	129.43	11.66	48.11	106.94	187.50	268.57

The table presents descriptive statistics for county-level crime rates, covering the sample period from 2005 to 2007. We report the number of observations, mean, standard deviation, 25th percentile, 50th percentile, and 75th percentile.

Table B2
Additional Controls for Crime Rates

	(1)	(2)	(3)	(4)	(5)	(6)
CPRATIO	0.0011** (0.0005)	0.0012** (0.0005)	0.0009** (0.0004)	0.0009** (0.0004)	0.0011** (0.0005)	0.0011** (0.0005)
Part I Crime	0.0000* (0.0000)				0.0000* (0.0000)	
Violent		-0.0000 (0.0000)				-0.0000* (0.0000)
Property		0.0000*** (0.0000)				0.0000*** (0.0000)
Arson		-0.0000 (0.0002)				-0.0001 (0.0002)
Part II Crime			0.0000 (0.0000)		-0.0000 (0.0000)	
Against person				0.0000 (0.0000)		0.0000 (0.0000)
Public disorder				0.0000 (0.0000)		0.0000 (0.0000)
Drug & alcohol				0.0000 (0.0000)		0.0000 (0.0000)
White collar				-0.0001* (0.0000)		-0.0001* (0.0000)
All others				-0.0000 (0.0000)		-0.0000 (0.0000)
Geographic Controls	Y	Y	Y	Y	Y	Y
Loan Chars Controls	Y	Y	Y	Y	Y	Y
Originator FE	Y	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y	Y
Half-year FE	Y	Y	Y	Y	Y	Y
Observations	2,364,231	2,364,231	2,384,686	2,384,686	2,364,231	2,364,231
Adj. R ²	0.318	0.318	0.318	0.318	0.318	0.318

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. In addition to the control variables in Table 3, we also include crime rates (per 10,000 residents per year) for Part I and Part II offenses, as well as their subcategories. All continuous variables, except for the crime rates, are winsorized at the 0.5 percent level and then standardized. All specifications include state and year fixed effects. White robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

C Geographic Variables Correlation

One concern with using local gambling preference is its potential correlation with other geographic variables. Therefore, we examine the correlation between CPRATIO and other geographic variables at the county-year level. Since some geographic variables are at more granular level, we do the following adjustments. Income and HPA at the county level are obtained from the same data sources as the zip code level data, which are the U.S. Census Bureau and the Federal Housing Finance Agency, respectively. Unemployment is the average of the monthly unemployment rate within each year.

Table C1 shows the correlation between geographic variables. None of the correlations in the first column exceed 0.32, indicating that CPRATIO contains information not solely covered by other common geographic characteristics. Therefore, the results from categorizing counties by CPRATIO are less likely driven by any other single geographic characteristic.

Table C1
Geographic Variables Correlation

	CPRATIO	REL	Income	Education	Married	Urban	Total population	Over65	Male-female ratio	Minority	Unemployment	HPA
CPRATIO	1.00											
REL	-0.01	1.00										
Income	0.26	-0.11	1.00									
Education	0.26	-0.10	0.64	1.00								
Married	-0.19	0.12	0.13	-0.30	1.00							
Urban	0.32	-0.01	0.41	0.52	-0.40	1.00						
Total population	0.32	-0.18	0.50	0.53	-0.37	0.74	1.00					
Over65	-0.15	0.31	-0.42	-0.34	0.23	-0.33	-0.37	1.00				
Male-female ratio	0.05	-0.21	0.12	-0.01	0.18	-0.17	-0.18	-0.21	1.00			
Minority	0.08	-0.09	-0.15	0.02	-0.52	0.20	0.23	-0.30	-0.12	1.00		
Unemployment	-0.06	-0.20	-0.34	-0.40	-0.14	-0.17	-0.08	0.02	-0.07	0.22	1.00	
HPA	0.26	-0.29	0.18	0.23	-0.08	0.15	0.23	-0.01	0.11	0.14	-0.21	1.00

The table shows the correlation between geographic variables at the county-year level. Income and HPA at the county level are obtained from the same data sources as the zipcode level data, which are the U.S. Census Bureau and the Federal Housing Finance Agency, respectively. Unemployment is the average of the monthly unemployment rate within each year.

D Diff-in-disc concerns

To validate our application of the diff-in-disc approach in comparing the number of loans and default rates between high and low gambling preference areas, we address the following concerns in this section.

D.1 Assumptions for Applying Diff-in-disc

According to Grembi et al. (2016), the validity of the diff-in-disc approach depends on three assumptions. First, all potential outcomes must be continuous around the cutoff. In our setting, this means no manipulation of FICO scores around the cutoff point of 620. Since the data on the distribution of FICO scores in the U.S. population is not available to us, we argue this assumption by referencing the findings in Keys et al. (2010). They found that the distribution of FICO scores across the population is smooth using data from an anonymous credit bureau. Additionally, by exploring the reversal of anti-predatory lending laws in Georgia and New Jersey, they found that borrowers were either unaware of the differential screening around the threshold or unable to quickly manipulate their FICO scores¹².

Second, in the absence of treatment, the effect of the confounding policy around the cutoff is constant over the "diff" part. To fulfill this assumption, we check whether the pattern holds for full documentation loans around FICO 620, which have no treatment but similar confounding factors. Table D1 shows the results for the number of loans (panel A) and default rate (panel B) estimated using full-documentation loans. For full-documentation loans, the number of loans shows a minimal jump, and the difference between high and low gambling preference areas is slight. This evidence indicates that without the ease of securitization, having a FICO score below or above 620 would not cause the number of loans to increase differently between high and low gambling preference areas. Additionally, there is no jump in the default rate for all types of loans, and the difference in the increase of the default rate between high and low gambling preference areas is minor. These findings illustrate that without the ease of securitization, lenders do not have lax screening standards for any type of loan and do not screen differently between high and low gambling preference areas.

Third, the effect of the treatment around the cutoff does not depend on the confounding policy. This assumption states that there should be no interaction between the treatment and the confounding policy. In other words, the pre-determined outcomes and covariates should have similar jumps (or no jumps) between high and low gambling preference areas. We test this assumption by estimating equation 6 with pre-determined outcomes and covariates as the

¹²According to the rating agency (Fair Isaac), strategic manipulation of FICO scores is difficult.

dependent variable and also add other control variables in the regression. We choose the same bandwidth (10) and kernel (uniform) as the main tests. The results are reported in Table D2. Column (1) presents the results of loan characteristics, showing that most characteristics have similar jumps between high and low gambling preference areas. The proportions of negative amortization and second-home loans are significant, but we argue that this does not affect our conclusion for several reasons. First, although the absolute jump magnitude is different, the ratio of 620^+ to 620^- is close, i.e., negative amortization jumps from 6.74/2.97 percent to 15.82/6.61 percent (2.35/2.23 times) in high/low gambling preference areas. Second, the proportion of these specific loans is too low to have a meaningful effect on the number of loans with misrepresentation and the default rate (i.e., in this local range, second homes make up only 1.2 percent of all loans and only 0.6 percent of misreported loans). Column (2) reports the results of borrower characteristics, showing that most characteristics have similar jumps between high and low gambling preference areas. The ratios of older people, minorities, and the male-female ratio are significant, but their influence is minor because the magnitude is too small, i.e., the average proportion of older people, the male-female ratio, and minorities in the local range are 12.2 percent, 28.89 percent, and 0.96 percent, while the differences are only 0.319 percent, -0.534 percent, and 0.003 percent, respectively.

D.2 Sensitivity to Regression Choices

Beyond these three assumptions, we also check whether the findings are sensitive to the inclusion of control variables, the choice of bandwidth, and the choice of estimation kernel. We estimate models with and without covariates, choosing bandwidths of 8, 10, and 12, and using uniform or triangular kernels. Table D3 reports the ratio of estimated values at FICO 620^+ to 620^- for the number of loans¹³. The magnitude of jumps is similar to the baseline results: the number of different types of loans increases significantly due to the ease of securitization, and the jump for loans with misreported seconds is smaller in high gambling preference areas than in low gambling preference areas. These results confirm the robustness of the findings that lenders do not play a larger role in increasing second-lien misrepresentation in high gambling preference areas. Table D4 shows that the default rate of loans with misreported seconds is small and insignificant in all cases. Moreover, although not significant, the jump in high gambling preference areas is smaller than the jump in low gambling preference areas in most cases. These robustness tests show that lenders in high gambling preference areas are not more inclined to increase such misrepresentation than lenders in low gambling preference areas.

¹³Since the number of loans is counted at the level of gambling preference areas while control variables are at the level of counties, control variables could not be included in the tests.

D.3 Placebo Test

Lastly, we conduct a placebo test to evaluate the possibility that our results are driven by chance. Specifically, following Goodman and Mayer (2018), we implement the diff-in-disc estimations at false FICO score thresholds below and above 620. Our inferences on the lender's role come from two findings from the diff-in-disc estimations: (1) the number of loans with misreported seconds increases much less in high gambling preference areas at FICO 620, and (2) the default rate of loans with misreported seconds either shows a small and insignificant jump in high gambling preference areas or a jump that is smaller in high gambling preference areas than in low gambling preference areas. Thus, at these false thresholds, we expect to find (1) no systematic difference in the increase of the number of loans with misreported seconds between high and low gambling preference areas as in our baseline results, and (2) no situation where the default rate of loans with misreported seconds shows a meaningful upward jump in high gambling preference areas and a jump that is larger in high gambling preference areas than in low gambling preference areas. To maintain a similar setting and avoid potential jumps in the misrepresentation rate found in the literature, we conduct the tests within the range of 601 to 639¹⁴. For the number of loans, we exclude the range of 615 to 625 to stay sufficiently away from the true threshold that could cause noise in jump estimation. Figure D1 plots the cumulative density function of the 28 placebo point estimates for the number of loans with misreported seconds. All of the placebo coefficients are greater than our estimated coefficient (-2.063) and smaller than the absolute value of our estimated coefficient with a small magnitude. We also perform a t-test on the placebo coefficients to check the null hypothesis that the true mean equals 0, and the p-value of this test is 0.819, which fails to reject the null hypothesis. These placebo tests support the idea that the difference in discontinuities for the number of loans with misreported seconds between high and low gambling preference areas is not due to chance. For the default rate, we only exclude the true threshold since no jump exists at the true threshold, so estimations at the scores near it would not be affected. Figure D2 plots the cumulative density function of the placebo point estimates for the default rate of loans with misreported seconds. Most placebo coefficients are smaller than the absolute value of our estimated coefficient (0.052). The larger ones are also insignificant, and they show an insignificant small jump in high gambling preference areas. The t-test on the placebo coefficients that checks whether the true mean equals 0 provides a p-value of 0.638. These placebo tests support the idea that lenders did not have laxer screening in high gambling preference areas.

¹⁴From Figure 3 in Griffin and Maturana (2016) about the misrepresentation rate, we can see clear jumps at FICO 600 and 640.

Table D1
Discontinuities of Full-documentation Loans Around the Credit Threshold

	(1) High	(2) Low	(3) All	(4) High-Low
Panel A. Number of loans (Est. 620+/620-)				
All loan	1.244	1.154	1.186	0.090 (2.27)
Reported second	1.183	1.152	1.161	0.031 (0.58)
Misreported second	1.325	1.216	1.251	0.109 (1.03)
Panel B. Default rate				
All loan	0.002 (0.26)	0.009 (0.23)	0.007 (0.49)	-0.007 (-0.56)
Reported second	-0.019 (-0.84)	0.004 (-0.01)	-0.004 (-0.41)	-0.022 (-0.67)
Misreported second	-0.015 (-0.84)	0.031 (0.26)	0.017 (-0.26)	-0.047 (-1.15)

The table presents the results of the estimations in Table 5 and Table 6 using full-documentation loans. Panel A reports the estimates for the number of loans at each FICO score. Panel B reports the estimates for loans that become 90 days or more delinquent using the MBA method in the first three years after origination. Using local linear regressions of the RDD approach, we estimate the number of loans and the default rate (with control variables) at FICO 620⁻ and FICO 620⁺ and compute the ratio of the estimated number of loans at FICO 620⁺ to 620⁻. Using the diff-in-disc approach, we estimate the difference between high and low gambling preference areas. We perform the estimation for all loans, loans with correctly reported simultaneous seconds, and loans with misreported simultaneous seconds. The results for loans in high gambling preference areas, loans in low gambling preference areas, loans in all areas, and the difference between high and low gambling preference areas are reported in columns (1) to (4), respectively. For discontinuities, bias-corrected t-values (standard errors clustered at the county level) following Calonico et al. (2014) are reported in parentheses. For the difference in the ratio of the number of loans/default rate between high and low gambling preference areas, t-statistics based on heteroskedasticity-consistent standard errors/standard errors clustered at the county level are reported in parentheses.

Table D2
Estimates of the discontinuities in observable covariates

	(1)		(2)
	loan characteristics		borrower characteristics
Interest rate (%)	0.001 (0.041)	REL	-0.005 (0.004)
Balance	0.017 (0.013)	Unemployment (%)	0.056 (0.071)
LTV (%)	0.128 (0.367)	HPA	-0.004 (0.004)
ARM	-0.011 (0.013)	Education (%)	-0.276 (0.242)
Negative amortization	0.025*** (0.007)	Married (%)	-0.052 (0.107)
Option ARM	0.001 (0.002)	Income	0.005 (0.007)
Prepayment penalty	0.004 (0.015)	Urban (%)	0.539 (0.335)
Cash-out	0.002 (0.013)	Total population (ln)	0.031 (0.025)
No-cash-out	0.002 (0.007)	Over65 (%)	0.319*** (0.105)
Investment	0.007 (0.007)	Male-female ratio	0.003** (0.001)
Second-home	0.007** (0.003)	Minority (%)	-0.534* (0.314)

The table reports the results of diff-in-disc regression for pre-determined outcomes and covariates using Equation 6 with other controls. The dependent variables are the pre-determined outcomes and covariates. Column (1) shows the tests for loan characteristics, and column (2) shows the tests for borrower characteristics. As in the main test, the bandwidth is set to 10 and the kernel is uniform. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D3
Sensitivity of Results for Number of Loans

	Uniform				Triangular			
	High	Low	All	H-L	High	Low	All	H-L
Bandwidth = 8								
All loan	2.230	2.242	2.237	-0.013 (-0.13)	2.250	2.204	2.225	0.046 (0.44)
Reported second	5.469	4.693	4.999	0.776 (0.96)	5.761	5.235	5.446	0.526 (0.56)
Misreported second	2.709	5.095	3.957	-2.385 (-2.85)	2.496	5.128	3.826	-2.632 (-3.21)
Bandwidth = 10								
All loan	2.240	2.289	2.266	-0.049 (-0.52)	2.240	2.234	2.237	0.006 (0.06)
Reported second	5.498	4.581	4.939	0.917 (1.23)	5.616	4.971	5.228	0.645 (0.74)
Misreported second	2.763	4.826	3.838	-2.063 (-2.83)	2.609	5.133	3.900	-2.524 (-3.20)
Bandwidth = 12								
All loan	2.199	2.245	2.224	-0.047 (-0.50)	2.237	2.262	2.250	-0.025 (-0.27)
Reported second	5.209	4.602	4.846	0.607 (0.91)	5.520	4.845	5.114	0.676 (0.84)
Misreported second	2.656	4.388	3.590	-1.732 (-2.60)	2.668	4.971	3.858	-2.302 (-3.20)
Bandwidth = 14								
All loan	2.197	2.199	2.198	-0.002 (-0.02)	2.222	2.257	2.241	-0.036 (-0.39)
Reported second	5.278	4.666	4.915	0.612 (0.95)	5.424	4.788	5.043	0.636 (0.84)
Misreported second	2.515	3.940	3.275	-1.426 (-1.99)	2.645	4.755	3.747	-2.110 (-3.18)

The table reports the sensitivity of RDD and diff-in-disc regression for the number of loans to the choice of bandwidth and estimation kernel. For the difference between high and low gambling preference areas, t-statistics based on heteroskedasticity-consistent standard errors are reported in parentheses.

Table D4
Sensitivity of Results for Default Rate of Loans with Misreported Second

	Uniform				Triangular			
	High	Low	All	H-L	High	Low	All	H-L
Bandwidth = 8								
No control	0.005 (-0.24)	0.019 (-0.83)	-0.004 (-1.18)	-0.014 (-0.23)	-0.007 (0.81)	-0.016 (-1.43)	-0.031 (-0.89)	0.009 (0.13)
With control	0.026 (-0.39)	0.048 (-1.27)	0.033 (-1.27)	-0.022 (-0.43)	-0.001 (0.45)	-0.012 (-1.73)	-0.007 (-0.97)	0.010 (0.16)
Bandwidth = 10								
No control	0.001 (-0.07)	0.052 (-0.86)	0.014 (-1.15)	-0.052 (-0.98)	-0.004 (0.32)	0.011 (-1.15)	-0.014 (-1.07)	-0.015 (-0.25)
With control	0.013 (0.16)	0.078 (-0.92)	0.045 (-0.73)	-0.065 (-1.41)	0.011 (0.06)	0.025 (-1.43)	0.016 (-1.06)	-0.014 (-0.26)
Bandwidth = 12								
No control	-0.022 (0.30)	0.065 (-0.30)	0.014 (-0.50)	-0.087 (-1.71)	-0.003 (0.09)	0.033 (-1.00)	0.000 (-1.14)	-0.036 (-0.65)
With control	-0.004 (0.33)	0.082 (0.07)	0.042 (0.15)	-0.085 (-1.95)	0.011 (0.12)	0.050 (-1.12)	0.030 (-0.82)	-0.038 (-0.80)
Bandwidth = 14								
No control	0.007 (-0.47)	0.071 (0.11)	0.033 (-0.72)	-0.064 (-1.39)	-0.007 (0.06)	0.046 (-0.59)	0.007 (-0.88)	-0.053 (-1.03)
With control	0.017 (-0.29)	0.081 (0.79)	0.051 (0.26)	-0.064 (-1.53)	0.008 (0.13)	0.060 (-0.43)	0.035 (-0.35)	-0.052 (-1.18)

The table reports the sensitivity of RDD and diff-in-disc regression for the default rate of loans with misreported seconds to the inclusion of control variables, the choice of bandwidth, and the choice of estimation kernel. For discontinuities, bias-corrected t-values (standard errors clustered at the county level) following Calonico et al. (2014) are reported in parentheses. For the difference between high and low gambling preference areas, t-statistics based on heteroskedasticity-consistent standard errors are reported in parentheses.

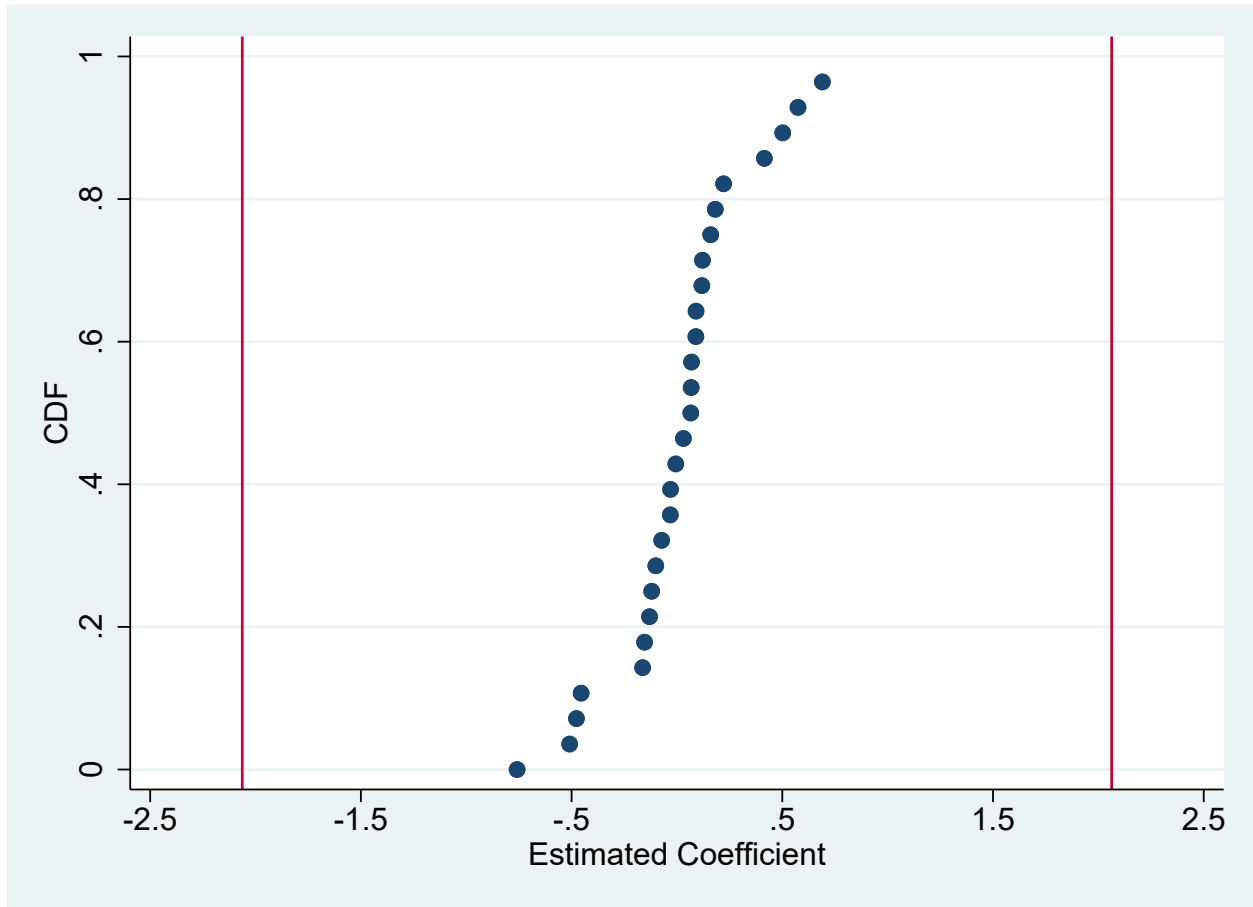


Figure D1

Placebo Tests for Number of Loans with Misreported Second

The figure plots the empirical c.d.f. of the estimated coefficient for the number of loans with misreported seconds from a set of diff-in-disc estimations at false FICO score thresholds below and above 620 (i.e., any score from 601 to 614 and any score from 626 to 639). The vertical lines show the benchmark estimate (-2.063) and its positive value (2.063).

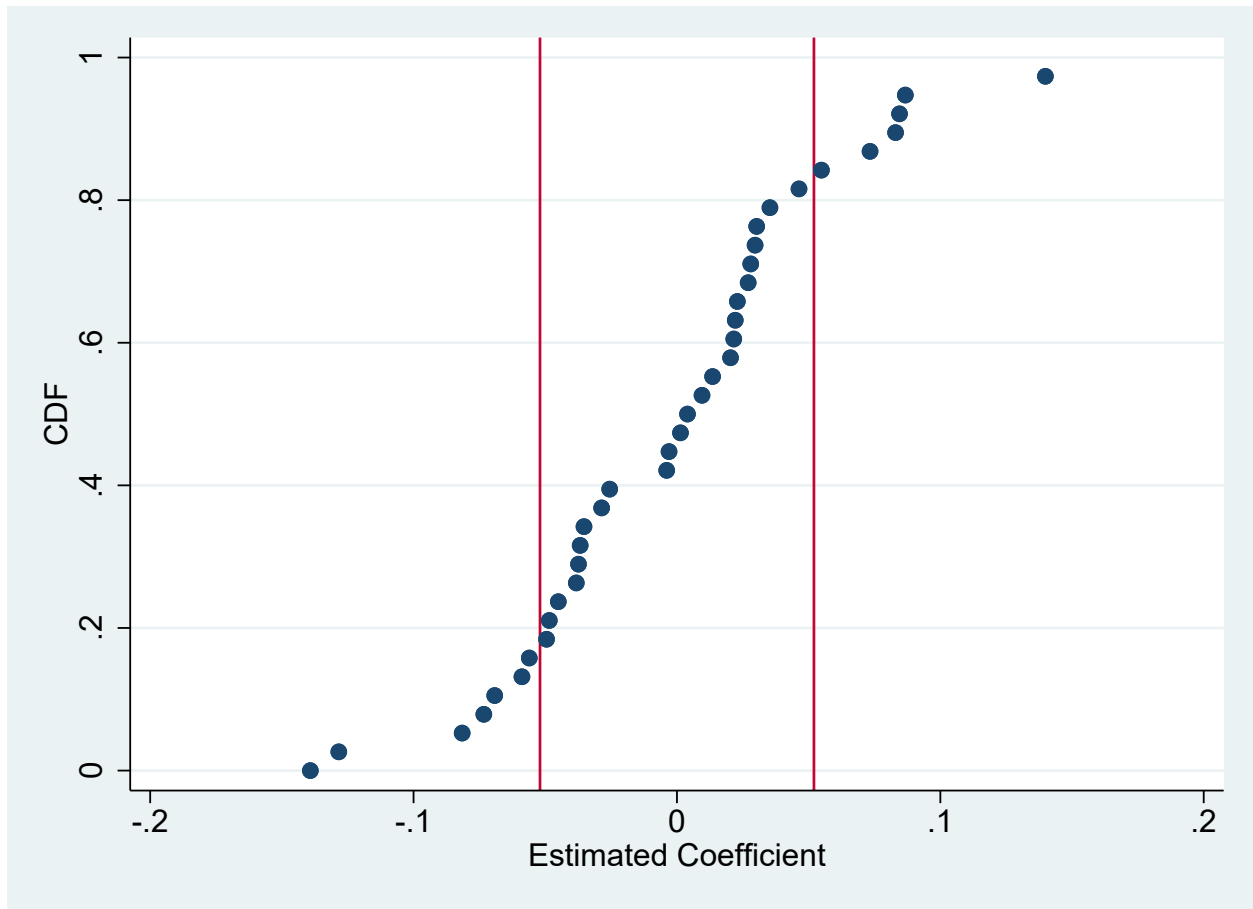


Figure D2

Placebo Tests for Default Rate of Loans with Misreported Second

The figure plots the empirical c.d.f. of the estimated coefficient for the default rate of loans with misreported seconds from a set of diff-in-disc estimations at false FICO score thresholds below and above 620. The vertical lines show the benchmark estimate (-0.052) and its positive value (0.052).